TAB 1: Capstone 1 main pages

INTRODUCTION

Project Context

The way information is managed continues to change as technology becomes more advanced, yet many school libraries still face challenges in organizing and maintaining their resources at STI College Alabang, the library serves a large number of students from different academic programs, each needing access to various books, references, and learning materials every day. Because of this high demand, the library needs a system that can efficiently track and manage all resources.

At present, the library still uses manual methods such as writing borrowed books in logbooks or recording them in spreadsheets. Librarians also check shelves manually to see if books are available. These methods take a lot of time and can lead to mistakes. Records may not always be updated right away, information can be misplaced, and handwriting may be difficult to read. Because of this, students sometimes have a hard time finding the books they need, and librarians struggle to keep accurate records.

This problem is not only seen in STI College Alabang but also in many schools that still use traditional library systems. As the number of students and books increases, it becomes harder to manage all transactions properly. Books may be returned late, misplaced, or even lost without proper tracking. Over time, this can lead to incorrect inventory records, which affects how the library operates.

To solve these problems, this study proposes a web-based library book reservation, borrowing, and inventory management system with QR-based scanning for STI College Alabang. The system is designed to make library transactions faster and easier. Students can reserve books online, while librarians can manage records through a single system. QR codes will be used to quickly identify books and process borrowing and returning.

With QR-based scanning, each book is given a unique code that can be scanned using a device. This makes the process faster and reduces human error. It also allows the system to update records in real time, helping ensure that the inventory is always accurate.

The system also aims to improve convenience for both students and librarians. Students will no longer need to manually search for book availability, as they can easily check and reserve materials through the system. Librarians, on the other hand, can spend less time on manual encoding and focus more on assisting students.

In addition, the system helps improve data safety and organization. Digital records are easier to store, update, and back up compared to manual records. It also helps reduce the risk of losing important information and ensures that only authorized users can access or modify data.

Lastly, the system can help the library make better decisions. With organized data, librarians can see which books are most used, which materials need updating, and how resources are being used overall. This helps improve the library’s services and better support student needs.

Purpose and Description

This project provides STI College Alabang with a streamlined digital library management system that replaces manual processes with efficient QR code scanning and online book reservations. Designed for reliability, the user-friendly platform centralizes library tasks into one accessible web interface, ensuring that data is managed effectively and operations remain consistent. By modernizing circulation and inventory tracking, the system enhances data security and provides a more organized, seamless experience for both students and Admin.

Statement of the Problem

General Problem

The primary challenge of this research is to design, develop, and implement a comprehensive, web-based library management ecosystem for STI College Alabang that replaces manual or fragmented processes with a unified digital framework. Specifically, the problem centers on how to effectively integrate Microsoft O365 authentication for secure institutional access and QR-based scanning technology for efficient asset tracking, while simultaneously providing a dual-interface system that automates the entire lifecycle of book reservations, borrowing transactions, and inventory auditing to ensure data integrity and operational efficiency for both students and administrators.

Specific Problem

Super Admin

How to implement a secure login module using Microsoft O365 authentication for Super Administrator?

In this module, Super Administrator can access the system using their institutional accounts to verify their identity. This ensures that only authorized personnel can access sensitive library administrative tools and data through a standardized sign-in process.

How to develop a centralized dashboard module that provides a real-time overview of daily library operations for super administrators?

In this module, super administrators can monitor key metrics such as daily visitors, registered users, total inventory, and active loans in real time. This ensures efficient resource tracking while providing quick-access shortcuts to core features like the circulation desk, user directories, and audit logs, alongside a visual gallery of recently cataloged library assets.real time

How to design an efficient inventory module that enables automated and batch cataloging operations for the super administrator?

In this module, the super administrator can manage the library's collection by automatically fetching book details via ISBN lookup and manually finalizing storage and classification records. This ensures streamlined data entry through batch spreadsheet imports, sequential bulk lookups, and fully customizable data export options for flexible catalog management.

How to automate the borrowing and returning process using QR-code scanning technologies for the circulation desk?

In this module, the system utilizes QR-code scanning of student IDs and book codes to automatically link borrowers to their selected items. This ensures a fast and accurate transaction process by significantly reducing manual data entry and minimizing administrative wait times.

How to create a comprehensive borrowing logs module that allows the super administrator to monitor and filter book transaction histories?

In this module, the super administrator can track library items through a centralized, view-only list displaying borrower identities, book information, and critical transaction timelines. This ensures efficient record tracking by utilizing status tags for overdue or active loans alongside built-in search and filter controls to quickly isolate specific data.

How to develop an analytics module with data visualization tools to track system trends and library performance for the super administrator?

In this module, the super administrator can filter performance metrics across dynamic timeframes to evaluate daily traffic, book circulation, asset statuses, and member registration. This ensures informed, data-driven decision-making by utilizing line graphs, status donut charts, and high-demand book leaderboards to visually present complex library operational trends.

How to implement a comprehensive user directory module that enables the super administrator to manage and modify registered user accounts?

In this module, the super administrator can view a master list of users detailing names, unique IDs, system roles, and academic programs. This ensures efficient administration by utilizing quick search and filter tools, alongside dedicated profile pages where administrators can update system access, modify account statuses, or archive profiles.

How to implement an attendance logs module using card-scanning technology to monitor student traffic within the library?

In this module, the super administrator can track library usage by checking students in and out using automated library card scans. This ensures accurate administrative monitoring by generating a complete real-time list of visitors paired with precise entry and exit timestamps to monitor daily traffic patterns efficiently.

How to develop a centralized settings and policies module that enables the super administrator to manage operational rules and backend configurations?

In this module, the super administrator can define circulation limits, organize library categories, configure academic programs, and publish system announcements. This ensures cohesive system management by combining operational policy controls with backend maintenance tools, troubleshooting logs, and automated communication scripts in a unified setup view.

How to implement a secure audit logs module that tracks, filters, and exports system activities for administrative accountability?

In this module, the super administrator can access a security tracking system that records timestamps, user identities, and technical differences for every action taken. This ensures rigorous system oversight by providing live-refresh data, advanced search and dropdown filters, and seamless data import/export capabilities in CSV and JSON formats

How to design an efficient notification module that alerts the super administrator about critical system events and updates in real time?

In this module, the super administrator can access recent alerts instantly through a quick drop-down panel in the system header or navigate to a dedicated notifications page. This ensures timely awareness and streamlined alert management by utilizing a search bar and quick filter tabs to organize complex system updates.

How to create a secure, view-only profile module that displays comprehensive account details for logged-in users?

In this module, authenticated users can review their own institutional and contact information including full name, system ID, official email, and security status on a read-only page. This ensures personal data transparency and role clarity by providing users with a standardized, tamper-proof overview of their account credentials and system privileges.

Librarian

How to implement a secure login module using Microsoft O365 authentication for Librarian?

In this module, Librarian can access the system using their institutional accounts to verify their identity. This ensures that only authorized personnel can access sensitive library administrative tools and data through a standardized sign-in process.

How to develop a centralized dashboard module that provides a real-time overview of daily library operations for the librarian?

In this module, the librarian can monitor critical real-time data such as daily visitor counts, registered users, total book inventory, and active loans for efficient resource tracking. This ensures streamlined workflow management by providing quick-access shortcuts to core features like user directories, audit logs, and analytics, alongside a visual showcase of recently cataloged library assets.

How to design an efficient inventory module that enables automated and batch cataloging operations for the librarian?

In this module, the librarian can manage the library's collection by automatically fetching book details via ISBN lookup and manually refining specific storage and classification records. This ensures optimal workflow efficiency through batch catalog panel tools, drag-and-drop spreadsheet imports, sequential bulk lookups, and fully customizable format exports.

How to automate the borrowing and returning process using QR-code scanning technologies for the librarian at the circulation desk?

In this module, the system utilizes QR-code scanning of student IDs and book codes to automatically link borrowers to their designated items. This ensures a swift transaction process for the librarian by drastically reducing manual text entry and minimizing administrative processing time.

How to create a centralized borrowing logs module that allows the librarian to track and filter historical book transactions?

In this module, the librarian can track loan lifecycles through a comprehensive, view-only list that showcases borrower identities, book data, and strict timelines for due dates and returns. This ensures quick record discovery by utilizing distinct status tags for active or overdue items coupled with targeted search fields and quick-filter buttons.

How to build an analytics module with dynamic visualization components to track library trends and system performance for the librarian?

In this module, the librarian can filter and evaluate system metrics across tailored timeframes to monitor visitor distribution, circulation volumes, and membership growth. This ensures intuitive insights by utilizing visual line graphs, asset allocation donut charts, and high-demand catalog leaderboards to break down complex operational data.

How to implement a structured user directory module that allows the librarian to review accounts and modify specific status permissions?

In this module, the librarian can oversee a comprehensive user register detailing names, unique student IDs, system roles, and academic programs. This ensures effective user management by integrating rapid search and filter systems alongside dedicated profile expansion pages where the librarian can review contact details and modify account statuses between active and pending states.

How to integrate an attendance logs module utilizing card-scanning technology to assist the librarian in tracking daily library traffic?

In this module, the librarian can monitor daily student facility utilization through automated library card scanning at entry and exit points. This ensures precise administrative tracking by populating an active visitor roster paired with accurate, real-time timestamps to analyze traffic density effectively.

How to configure a dedicated settings and policies module that enables the librarian to adjust library rules and manage student support logs?

In this module, the librarian can utilize several specialized views to establish circulation boundaries, arrange book categories, align academic paths, and push wide-reaching system announcements. This ensures cohesive administrative management by providing an accessible Q&A support environment tailored to troubleshooting and assisting students.

How to construct an intuitive notification module that keeps the librarian instantly updated on major system actions and changes?

In this module, the librarian can view urgent alerts dynamically via a quick-drop header widget or transition to a standalone notifications control page. This ensures rapid awareness and response times by using intuitive filter tabs and specialized keyword search inputs to sort through system-generated event updates.

How to establish a secure, read-only profile module for librarians to safely verify their personal account credentials?

In this module, the logged-in librarian can cross-reference their account specifics including name, institutional ID, official email, academic programs, and security flags on a protected, view-only layout. This ensures system transparency by offering a clear, unalterable view of the user's specific registration record and clearance standing.

Student Assistant

How to implement a secure login module using Microsoft O365 authentication for Student Assistants?

In this module, Student Assistants can access the system using their institutional accounts to verify their identity. This ensures that only authorized personnel can access sensitive library administrative tools and data through a standardized sign-in process.

How to develop a centralized dashboard module that provides a quick overview of daily activities and personal metrics for student assistants?

In this module, student assistants can view real-time statistics for daily visitors, total users, and active loans, alongside quick navigation shortcuts to the circulation desk, attendance logs, and book catalog. This ensures dual-purpose tracking by also displaying a personalized bottom row showcasing the student assistant's current borrows, reservations, and recent attendance.

How to design an intuitive catalog module that allows student assistants to browse, search, and reserve items from the book collection?

In this module, student assistants can look up books using a keyword search bar for titles, authors, or ISBN codes, and refine results using dropdown filters for categories and availability indicators. This ensures a streamlined discovery process while enabling users to secure items directly via an integrated reservation button.

How to implement an inventory monitoring module with tabular data and QR-code generation for student assistants?

In this module, student assistants can monitor registered book items using a structured data table displaying titles, authors, categories, and ISBN codes. This ensures easy tracking and item location through category selection filters, search options, and individual system-generated QR codes for every book item.

How to automate the borrowing and returning process using QR-code scanning technologies for the student assistant at the circulation desk?

In this module, the system utilizes QR-code scanning of student IDs and book codes to automatically link borrowers to their selected items. This ensures a swift transaction process for the student assistant by minimizing manual data entry and reducing wait times at the desk.

How to create a structured borrowing history module for student assistants to track their personal book transactions?

In this module, student assistants can view their personal transaction timelines, including borrowed dates, due dates, actual return dates, and item statuses. This ensures easy tracking of personal historical data by utilizing status filters and a specialized timeline search bar.

How to implement a view-only borrowing logs module that allows student assistants to monitor global library transactions?

In this module, student assistants can access a centralized list tracking borrower details, book titles, and due-date timelines across the library. This ensures simple record verification through clear status tags alongside built-in search and filtering tools to quickly locate specific logs.

How to design a personal attendance tracking module for student assistants to review their library visitation history?

In this module, student assistants can reference a structured data table showing dates, check-in/check-out timestamps, and session statuses for their library visits. This ensures transparent personal record keeping by allowing users to seamlessly review their chronological logs through an all-time history option.

How to integrate an attendance logs module using card-scanning technology to help student assistants track daily library traffic?

In this module, student assistants can track facility utilization by scanning student library cards at entry and exit points. This ensures straightforward administrative monitoring by populating a complete, real-time list of checked-in and checked-out visitors with precise timestamps.

How to build a responsive notification module that updates student assistants on system-wide events and alerts?

In this module, student assistants can view urgent updates via a header dropdown panel or manage comprehensive alerts on a dedicated notifications page. This ensures streamlined event monitoring through the use of quick filter tabs and targeted keyword search functionalities.

How to establish a secure, view-only profile module for student assistants to verify their personal account details?

In this module, logged-in student assistants can cross-reference their account information such as full name, system ID, account status, and membership role on a read-only page. This ensures personal data clarity and account security by providing a standardized, unalterable overview of their credentials.

Student

How to implement a secure login module using Microsoft O365 authentication for Students?

In this module, Students can access the system using their institutional accounts to verify their identity. This ensures that only authorized personnel can access sensitive library administrative tools and data through a standardized sign-in process.

How to design a personalized dashboard module that provides students with a real-time overview of their library account status and credentials?

In this module, students can access a digital library card with a scannable QR code alongside tracked sections for current borrows, active reservations, and recent attendance histories. This ensures a centralized, user-centric experience by also integrating an easily accessible support and FAQs panel for quick system troubleshooting.

How to implement an intuitive category browsing module that enables students to search, filter, and reserve books from the library collection?

In this module, students can explore books using specialized fractional availability indicators and search parameters for titles, authors, or ISBN codes. This ensures seamless book discovery and acquisition by pairing dynamic category and availability dropdown filters with a direct reservation hold button.

How to build a structured borrowing history module that allows students to monitor personal book transactions and due dates?

In this module, students can navigate a transactional table detailing library resources, borrowing dates, and strict timeline intervals for returns. This ensures efficient tracking of book statuses by utilizing a keyword timeline search bar and status filters to quickly isolate active, returned, or overdue materials.

How to develop a personal attendance tracking module for students to review their chronological library visits?

In this module, students can access a structured data table that displays precise dates, check-in/out timestamps, and session statuses for each library visit. This ensures clear personal visibility into facility usage by allowing users to toggle an all-time history option for full chronological tracking.

How to create a responsive notification module that alerts students regarding important library updates and usage events?

In this module, students can open a quick drop-down panel in the system header to view recent notifications or navigate to a dedicated page for more comprehensive tracking. This ensures that students stay informed of crucial status changes, library usage alerts, and upcoming transaction dates through clean filter categories.

How to establish a secure, view-only profile module for students to verify their institutional account information?

In this module, logged-in students can cross-reference their personal profiles, including full names, unique system IDs, program paths, and security configurations, on a read-only page. This ensures data transparency and identity verification by presenting an unalterable overview of the student's registration details.

OBJECTIVES

General Objectives

To design and develop Lumina, a web-based library book reservation, borrowing, and inventory management system with QR-based scanning for STI College Alabang. The system aims to enhance efficiency in library transactions through a centralized online platform, improve inventory tracking accuracy, and provide seamless real-time access to library services for both students and librarians.

Specific Objectives

Super Admin

To integrate a secure login module using Microsoft O365 authentication for identity verification.

This objective ensures a secure and standardized sign-in process by leveraging institutional accounts, verifying that only authorized personnel can access administrative tools.

To develop a Centralized Dashboard Module that displays a real-time overview of daily library operations, critical system metrics, and quick-access navigation shortcuts.

This objective aims to provide super admin with a centralized, high-level control panel of daily visitor counts, registered users, total book inventory, and active loans, enabling efficient resource monitoring, rapid system navigation, and data-driven management.

To develop an Inventory Module that enables Super administrators to manage the library’s book collection through automated ISBN fetching and detailed manual cataloging.

This objective aims to provide library staff with a streamlined system to add new items by automatically retrieving book information via ISBN, while allowing manual inputs for Dewey Decimal numbers, categories, copies, and exact shelf locations, enabling efficient asset tracking and organized resource management.

To develop a Circulation Desk Module that automates the library's borrowing and returning processes utilizing QR-code scanning technologies.

This objective aims to streamline front-desk operations by allowing staff to scan student IDs and book codes to automatically link borrowers to their items, significantly reducing manual data entry and transaction time while improving overall operational efficiency.

To develop a Borrowing Logs Module that provides a centralized, view-only tracking system for all book borrowing and returning transactions.

This objective aims to provide the super administrator and staff with a consolidated overview of key transaction details including borrower information, book metadata, transaction timelines, and real-time status tags integrated with search and filtering capabilities to enable rapid record retrieval and efficient monitoring of overdue assets.

To develop an Analytics Module that provides a visual data dashboard to track system trends, library performance, and historical data distributions.

This objective aims to provide the super administrator with key metric summaries including visitor traffic, active borrows, inventory volumes, and membership counts equipped with custom timeframe filters, line graphs, donut charts, and demand leaderboards to enable data-driven insights and efficient long-term resource planning.

To develop a User Directory Module that allows the Super Administrator to view, manage, and modify all registered user accounts in a centralized master list.

This objective aims to provide the Super Administrator with efficient user management capabilities including search and filter functionalities, detailed profile viewing, and administrative control tools to facilitate rapid account updates, role adjustments, status changes, and profile archiving.

To develop an Attendance Logs Module that enables the super administrator to monitor and track student traffic within the library in real time.

This objective aims to provide the super administrator with an automated tracking system that utilizes library card scanning to record student check-ins and check-outs alongside exact entry and exit timestamps, providing a streamlined method to analyze daily library usage and monitor visitor flow.

To develop a Setting and Policies Module that enables the super administrator to manage system rules, operational configurations, and backend maintenance through dedicated setup views.

This objective aims to provide the super administrator with centralized administrative controls including tools to define circulation limits, configure academic programs, organize book categories, and post announcements integrated with support logs, batch account resets, and automated email dispatches to ensure seamless

system customization and efficient backend troubleshooting.

To develop an Audit Logs Module that provides a secure tracking system to record, monitor, and review all administrative activities and system changes.

This objective aims to provide the super administrator with a comprehensive security table detailing exact timestamps, user identities, action types, affected entities, and technical differences, integrated with advanced search, dropdown filtering, live refreshing, and CSV export capabilities to ensure robust accountability, rapid incident investigation, and efficient compliance auditing.

To develop a Notification Module that alerts the super administrator about critical system events and updates via real-time header indicators and a dedicated management interface.

This objective aims to provide administrators and staff with immediate situational awareness through a header notification bell and drop-down panel, integrated with quick filter tabs and search bar capabilities on a dedicated alerts page to facilitate rapid event monitoring and efficient system update tracking.

To develop a Profile Module that provides a secure, view-only interface for logged-in users to access and review their personal account information.

This objective aims to provide users with a centralized, read-only profile page displaying critical account data including full names, system IDs, official emails, contact details, residential addresses, and membership statuses ensuring transparent data access, account clarity, and straightforward user detail verification.

Librarian

To develop a secure Login Module that serves as a standardized entry point for authorized library staff using Microsoft O365 authentication.

This objective aims to protect library data by restricting access to administrative tools, ensuring that only verified personnel can log in and manage the system's core backend features.

To develop a Centralized Dashboard Module that displays a real-time overview of daily library operations, critical system metrics, and quick-access navigation shortcuts.

This objective aims to provide Librarians with a centralized, high-level control panel of daily visitor counts, registered users, total book inventory, and active loans, enabling efficient resource monitoring, rapid system navigation, and immediate visibility of newly cataloged assets.

To develop an Inventory Module that enables Librarians to manage the library’s book collection through automated ISBN fetching and detailed manual cataloging.

This objective aims to provide library staff with a streamlined system to add new items by automatically retrieving book information via ISBN, while allowing manual inputs for Dewey Decimal numbers, categories, copies, and exact shelf locations, enabling efficient asset tracking and organized resource management.

To develop a Circulation Desk Module that automates the library's borrowing and returning processes utilizing QR-code scanning technologies.

This objective aims to streamline front-desk operations by allowing staff to scan student IDs and book codes to automatically link borrowers to their items, significantly reducing manual data entry and transaction time while improving overall operational efficiency.

To develop a Borrowing Logs Module that provides a centralized, view-only tracking system for all book borrowing and returning transactions.

This objective aims to provide the Librarian and staff with a consolidated overview of key transaction details including borrower information, book metadata, transaction timelines, and real-time status tags integrated with search and filtering capabilities to enable rapid record retrieval and efficient monitoring of overdue assets.

To develop an Analytics Module that provides a visual data dashboard to track system trends, library performance, and historical data distributions.

This objective aims to provide the super administrator with key metric summaries including visitor traffic, active borrows, inventory volumes, and membership counts equipped with custom timeframe filters, line graphs, donut charts, and demand leaderboards to enable data-driven insights and efficient long-term resource planning.

To develop a User Directory Module that allows the Librarian to view, manage, and modify registered user accounts from a centralized master list.

This objective aims to provide the Librarian with comprehensive user management capabilities including profile search and filtering to open detailed user profiles, review contact details, and update critical account parameters such as setting account statuses to Active or Pending.

To develop an Attendance Logs Module that enables the Librarian to monitor and track student traffic within the library in real time.

This objective aims to provide an automated tracking system that utilizes library card scanning to record student check-ins and check-outs alongside exact entry and exit timestamps, providing a streamlined method to analyze daily library usage and monitor visitor flow.

To develop a Setting and Policies Module that enables the Librarian to configure library rules, operational setups, and support documentation through dedicated views.

This objective aims to provide the Librarian with centralized administrative control to define circulation limits, set up academic programs, organize book categories, and broadcast system-wide announcements, while hosting a Q&A support log to assist students with system troubleshooting.

To develop a Notification Module that alerts the Librarian about critical system events and updates via real-time header indicators and a dedicated management interface.

This objective aims to provide administrators and staff with immediate situational awareness through a header notification bell and drop-down panel, integrated with quick filter tabs and search bar capabilities on a dedicated alerts page to facilitate rapid event monitoring and efficient system update tracking.

To develop a Profile Module that provides a secure, view-only interface for logged-in users to access and review their personal account information.

This objective aims to provide users with a centralized, read-only profile page displaying critical account data including full names, system IDs, official emails, contact details, residential addresses, and membership statuses ensuring transparent data access, account clarity, and straightforward user detail verification.

Student Assistant

To develop a secure Login Module that serves as a standardized entry point for library staff utilizing Microsoft O365 authentication.

This objective aims to safeguard system data by restricting administrative access to authorized personnel, ensuring that only verified staff members can access and manage library resources.

To develop a Centralized Dashboard Module that provides Student Assistants with a central control panel displaying daily activities, system metrics, and personal activity rows.

This objective aims to provide Student Assistants with immediate visibility into real-time metrics including daily visitors, total users, and active loans complemented by quick-access navigation shortcuts and a personalized tracking section for their own borrows, reservations, and recent attendance.

To develop a Catalog Module that enables Student Assistants to browse, search, and filter the library's entire book collection.

This objective aims to provide an interactive search interface displaying detailed book cards with fractional availability indicators, integrated with multi-parameter filtering options and a reserve for pickup function to allow users to efficiently locate and hold books.

To develop an Inventory Module that provides Student Assistants with a structured tabular overview to monitor and manage individual book copies within the system.

This objective aims to provide an organized data table detailing book titles, authors, categories, and ISBN codes, equipped with keyword search tools, category filters, and generated QR codes for every book item to streamline asset tracking.

To develop a Circulation Desk Module that automates the library's borrowing and returning processes utilizing QR-code scanning technologies.

This objective aims to streamline front-desk operations by allowing Student Assistants to scan student IDs and book codes to automatically link borrowers to their items, significantly reducing manual data entry and transaction time.

To develop a Borrowing Module that provides a structured timeline view of a Student Assistant's personal borrow history.

This objective aims to deliver an organized transactional table showing resource metadata, borrow dates, due date timelines, and status logs, embedded with search bars and status filters to let individuals monitor their active, returned, and overdue books.

To develop a Borrowing Logs Module that provides Student Assistants with a simple, view-only list to track broad system-wide borrowing and returning histories.

This objective aims to provide a centralized monitoring page displaying borrower data, book details, and transaction timelines alongside color-coded status tags, keyword search tools, and quick filters to easily find specific circulation records.

To develop an Attendance Module that provides students with a personal, read-only tracking record of their individual library visits.

This objective aims to grant students transparent access to their own check-in and check-out sessions, displaying exact dates, entry/exit timestamps, and session statuses through an all-time chronological history table.

To develop an Attendance Logs Module that enables Student Assistants to monitor and track student traffic within the library.

This objective aims to provide an automated tracking system that utilizes library card scanning to display real-time check-in and check-out logs alongside precise entry and exit timestamps, simplifying the process of evaluating daily library utilization.

To develop a Notification Module that alerts Student Assistants about important system events and updates through multi-layered interface elements.

This objective aims to maintain prompt situational awareness through a header notification bell and drop-down alert panel, paired with a dedicated alerts management page featuring search bars and filter tabs

for quick update navigation.

To develop a Profile Module that provides a secure, view-only interface for logged-in users to review their personal account information.

This objective aims to provide users with a clean, read-only display of critical account data including full names, system IDs, official emails, contact details, residential addresses, and membership roles ensuring transparent data verification and account clarity.

Student

To develop a secure Login Module that enables students to sign in using Microsoft O365 institutional accounts and includes a multi-step profile completion workflow.

This objective aims to ensure only verified institutional individuals gain entry, while requiring new users to submit residential, contact, and academic details for a pending administrative review state before full system access is granted.

To develop a Centralized Dashboard Module that provides students with a personalized, real-time overview of their library account status and support resources.

This objective aims to present students with an interactive interface featuring an exportable digital Library Card with a scannable QR code, real-time tracking tables for active borrows, reservations, and recent attendance durations, alongside a dedicated Support & FAQs panel.

To develop a Category Module that enables students to browse, look up, and place holds on items within the library's book collection.

This objective aims to provide students with an interactive visual catalog displaying detailed book cards and fractional availability indicators, integrated with multi-parameter search bars and dropdown filters, alongside a reserve for pickup feature.

To develop a Borrow Module that provides students with a structured, chronological timeline of their personal borrowing histories.

This objective aims to provide students with a comprehensive transactional table detailing resource metadata, borrow dates, and return timelines, equipped with keyword search bars and status filters to easily monitor active, returned, and overdue books.

To develop an Attendance Module that provides students with a detailed, personal record of their library check-in and check-out sessions.

This objective aims to give students transparent visibility into their daily library visits through a structured data table displaying exact timestamps and session statuses, complete with an all-time chronological history tracking option.

To develop a Notification Module that alerts students about critical library updates, system events, and personal usage logs.

This objective aims to maintain prompt situational awareness for students through a quick header notification bell and drop-down alert panel, paired with a dedicated notification page for managing and tracking library updates.

To develop a Profile Module that provides a secure, view-only interface for logged-in users to review their personal account information.

This objective aims to provide students with a clean, read-only display of their critical account data including full names, system IDs, official emails, contact numbers, residential addresses, and membership roles ensuring transparent data verification and account clarity.

Scope and Limitations

Technology has changed the way libraries work, making it much faster to track books and share information compared to old paper methods. Instead of searching through boxes of cards, computers now let us check if a book is available in just a few seconds. This project follows that trend by bringing digital tools to the STI College Alabang library.

Scope of the Study

Super Admin

Login Module

This module provides a secure and standardized entry point for library staff using Microsoft O365 authentication. It restricts access to administrative tools to ensure that only authorized personnel can manage library data.

Dashboard Module

This module provides a centralized, high-level control panel that gives Super administrators a real-time overview of daily operations and system metrics. It displays critical data including daily visitor counts, registered users, total book inventory, and active loans, allowing for efficient resource monitoring. Additionally, the module features quick-access navigation shortcuts to core administrative features such as circulation desk, book inventory, user directories, visitor logs, audit logs, and analytics and showcases a visual gallery of the most recently cataloged library assets.

Inventory Module

This module allows the Super Administrator to manage the library's collection of books by automatically fetching details via ISBN lookup and manually finalizing records like Dewey Decimal numbers, categories, copy counts, and shelf locations. It also includes a Batch Catalog Operations panel featuring an Import Sheet tool for uploading spreadsheets via drag-and-drop, an ISBN Fallback text field for bulk sequential lookups and cover art retrieval, and an Export Data manager with fully customizable column selection to download the active catalog as a Spreadsheet, CSV, JSON Data, or Covers ZIP file.

Circulation Desk Module

This module automates the borrowing and returning process by utilizing QR-Code scanning technologies. By scanning student IDs and book codes, the system automatically links the borrower to the item, reducing manual data entry and transaction time.

Borrowing Logs Module

This module shows a simple, view-only list that lets the super administrator track when books are borrowed and returned. It displays all the key details in one place, including the borrower's name and ID, the book title and author, the date the book was taken, and the timeline for due dates and return dates. It also uses clear status tags to show if a book is still out, returned, or overdue. Staff can easily use the search bar or click the filter buttons (All, Active, Returned, Overdue) to quickly find specific records.

Analytics Module

This module provides super administrator with a visual data dashboard to track system trends and performance over time. It allows users to filter data by specific timeframes, including 7 days, 30 days, or 1 year. The module features key metric summaries for daily visitor traffic, active book borrows, total inventory

volumes, and registered library members. It uses line graphs to display daily traffic distribution and circulation trends, a donut chart to break down asset allocation by status (returned, active, or overdue), and a leaderboard list showcasing the highest-demand catalog books based on borrow counts.

User Directory Modules

This module allows the Super Administrator to view and manage all registered user accounts in the system. It displays a master list of users with their names, emails, unique IDs, system roles, account statuses, and academic programs. Super Administrator can quickly search for accounts by name or email, or use the filter button to narrow down the list. Clicking on a user opens their detailed profile page, where super administrator can view contact information and update account settings, such as changing their system role, modifying their account status, or archiving the profile.

Attendance Logs Module

This module allows the super adminintrator to monitor and track student traffic within the library. By scanning the student's library card, the super administrator can see a complete list of all checked-in and checked-out students alongside the exact timestamp of their entry and exit, providing a simple way to track daily library usage.

Setting and Policies Modules

This module allows super administrator to manage system rules and operational configurations through several dedicated setup views. It enables super administrator to define circulation and reservation limits, configure student academic programs, organize book categories, and post system-wide announcements. Additionally, it contains a question-and-answer support log for troubleshooting, alongside backend maintenance tools for handling batch account resets, automated email dispatches, and configuration testing.

Audit Logs Module

This module provides a security tracking system that records all changes and activities made within the system. It displays a table detailing the exact timestamp of each action, the name of the administrator or user who performed it, the type of action taken, the affected entity, a short reason or description of the change, and a button to view specific technical differences. Super Admininstrator can easily manage large lists of logs using the search bar to find names, actions, or reasons, use dropdown filters to sort by specific entities or actions, refresh the data live, and Import and export data via Spreadsheet CSV and developer JSON.

Notification Module

This module alerts super Administrator about important system events and updates. Clicking the notification bell in the system header opens a quick drop-down panel showing recent alerts. The dedicated notifications page allows staff to manage alerts using quick filter tabs and find specific updates using a search bar to filter events.

My Profile Module

This module provides a view-only page where logged-in users can check their personal account details. It displays the user's full name, system ID, official email, contact number, residential address, account status, membership role, and security status.

Librarian

Login Module

This module provides a secure and standardized entry point for library staff using Microsoft O365 authentication. It restricts access to administrative tools to ensure that only authorized personnel can manage library data.

Dashboard Module

This module provides a centralized, high-level control panel that gives Librarian a real-time overview of daily operations and system metrics. It displays critical data including daily visitor counts, registered users, total book inventory, and active loans, allowing for efficient resource monitoring. Additionally, the module features quick-access navigation shortcuts to core administrative features such as user directories, audit logs, and analytics and showcases a visual gallery of the most recently cataloged library assets.

Inventory Module

This module allows the Super Administrator to manage the library's collection of books by automatically fetching details via ISBN lookup and manually finalizing records like Dewey Decimal numbers, categories, copy counts, and shelf locations. It also includes a Batch Catalog Operations panel featuring an Import Sheet tool for uploading spreadsheets via drag-and-drop, an ISBN Fallback text field for bulk sequential lookups and cover art retrieval, and an Export Data manager with fully customizable column selection to download the active catalog as a Spreadsheet, CSV, JSON Data, or Covers ZIP file.

Circulation Desk Module

This module automates the borrowing and returning process by utilizing QR-Code scanning technologies. By scanning student IDs and book codes, the system automatically links the borrower to the item, reducing manual data entry and transaction time.

Borrowing Logs Module

This module shows a simple, view-only list that lets the Librarian track when books are borrowed and returned. It displays all the key details in one place, including the borrower's name and ID, the book title and author, the date the book was taken, and the timeline for due dates and return dates. It also uses clear status tags to show if a book is still out, returned, or overdue. Staff can easily use the search bar or click the filter buttons (All, Active, Returned, Overdue) to quickly find specific records.

Analytics Module

This module provides super administrator with a visual data dashboard to track system trends and performance over time. It allows users to filter data by specific timeframes, including 7 days, 30 days, or 1 year. The module features key metric summaries for daily visitor traffic, active book borrows, total inventory volumes, and registered library members. It uses line graphs to display daily traffic distribution and circulation trends, a donut chart to break down asset allocation by status (returned, active, or overdue), and a leaderboard list showcasing the highest-demand catalog books based on borrow counts.

User Directory Module

This module allows the Librarian to view and manage all registered user accounts in the system. It displays a master list of users with their names, emails, unique IDs, system roles, account statuses, and academic programs. Librarian can quickly search for accounts by name or email, or use the filter button to narrow down the list. Clicking on a user opens their detailed profile page, where Librarian can view contact information and update account settings, modifying their account status to Active and Pending,

Attendance Logs Module

This module allows the Librarian to monitor and track student traffic within the library. By scanning the student's library card, the Librarian can see a complete list of all checked-in and checked-out students alongside the exact timestamp of their entry and exit, providing a simple way to track daily library usage.

Setting and Policies Module

This module allows Librarian to manage system rules and operational configurations through several dedicated setup views. It enables the Librarian to define circulation and reservation limits, configure student academic programs, organize book categories, and post system-wide announcements. Additionally, it contains a question-and-answer support log for helping students.

Notification module

This module alerts Librarian about important system events and updates. Clicking the notification bell in the system header opens a quick drop-down panel showing recent alerts. The dedicated notifications page allows Librarian to manage alerts using quick filter tabs and find specific updates using a search bar to filter events.

My Profile Module

This module provides a view-only page where logged-in users can check their personal account details. It displays the user's full name, system ID, official email, contact number, residential address, account status, membership role, and security status.

Student Assistants

Login Module

This module provides a secure and standardized entry point for library staff using Microsoft O365 authentication. It restricts access to administrative tools to ensure that only authorized personnel can manage library data.

Dashboard Module

This module provides Student Assistant with a central control panel that gives a quick overview of daily activities and system statistics. At the top, it displays real-time numbers for today's visitors, total users, and active book loans. Below these stats, it features quick system shortcuts for easy navigation to the Circulation Desk, Attendance Logs, and Book Catalog. Finally, the bottom section includes a visual row showing the student assistant his current borrows, Reservation, and Recent Attendance .

Catalog Module

This module provides Student Assistant to browse and search the library's book collection. Each book card displays the book's title, author name, cover image, and a fractional availability indicator showing the number of available copies. Users can look up items using a keyword search bar that supports titles, authors, or ISBN codes, and refine the list using dropdown filters for book categories, availability status, and alphabetical sorting. Additionally, each book contains a Reserve for Pickup button, allowing users to place a hold on their selected book.

Inventory Module

This module provides Student Assistants with a tabular overview to monitor and manage individual book copies within the system. The interface features a structured data table displaying the Title, Author, Category, and ISBN code for each registered book item. Student Assistants can filter the displayed inventory items using a category selection menu or quickly locate specific entries by typing keywords into the search bar. Additionally, the module includes a generated QR code for every book item.

Circulation Desk Module

This module automates the borrowing and returning process by utilizing QR-Code scanning technologies. By scanning student IDs and book codes, the system automatically links the borrower to the item, reducing manual data entry and transaction time.

My Borrowing Module

This module provides student assistant with a structured timeline of their borrowed history. The main interface features a transactional table listing the library Resource (displaying the title, and author), the Borrowed date, a detailed Timeline showing the due date along with actual return dates, and the current Status. Student Assistant can locate specific entries using a keyword timeline search bar or instantly filter the list by four distinct status views: All, Active, Returned, and Overdue.

Borrowing Logs Module

This module shows a simple, view-only list that lets the Student Assistant track when books are borrowed and returned. It displays all the key details in one place, including the borrower's name and ID, the book title and author, the date the book was taken, and the timeline for due dates and return dates. It also uses clear status tags to show if a book is still out, returned, or overdue. Staff can easily use the search bar or click the filter buttons (All, Active, Returned, Overdue) to quickly find specific records.

My Attendance Module

This module provides Students with a personal record of their library check-in and check-out sessions. The interface features a structured data table displaying the Date, Check In time, Check Out time, and session Status for each visit. Students can also view their chronological tracking logs through an All-Time History option

Attendance Logs Module

This module allows the Student Assistant to monitor and track student traffic within the library. By scanning the student's library card, the Student Assistant can see a complete list of all checked-in and checked-out students alongside the exact timestamp of their entry and exit, providing a simple way to track daily library usage.

Notification Module

This module alerts Student Assistant about important system events and updates. Clicking the notification bell in the system header opens a quick drop-down panel showing recent alerts. The dedicated notifications page allows Student Assistant to manage alerts using quick filter tabs and find specific updates using a search bar to filter events.

My Profile Module

This module provides a view-only page where logged-in users can check their personal account details. It displays the user's full name, system ID, official email, contact number, residential address, account status, membership role, and security status.

Student

Login Module

This module enables students to access the system securely by logging in with their Microsoft O365 institutional accounts, ensuring only verified individuals from the institution can sign in. For new users, the module features a profile completion step where they must provide their residential address, contact details, and academic program. Upon submission, the account enters a "Pending" state, restricting system utilization until a super administrator reviews and approves the profile.

Dashboard Module

This module provides students with a personalized overview of their library account status. The top section features their digital Library Card, showing their profile picture, name, program, academic year, a scannable QR code, and Back and Export actions. Below, the interface tracks three sections: Current Borrows (active

books and due dates), Reservations (books ready for pickup or queue placement with a cancel option), and Recent Attendance (a snapshot of check-in/out timestamps and visit durations). The bottom includes a Support & FAQs panel for students to know about the system.

Category Module

This module provides Student Assistant to browse and search the library's book collection. Each book card displays the book's title, author name, cover image, and a fractional availability indicator showing the number of available copies. Users can look up items using a keyword search bar that supports titles, authors, or ISBN codes, and refine the list using dropdown filters for book categories, availability status, and alphabetical sorting. Additionally, each book contains a Reserve for Pickup button, allowing users to place a hold on their selected book.

My Borrow Module

This module provides students with a structured timeline of their borrowed history. The main interface features a transactional table listing the library Resource (displaying the title, and author), the Borrowed date, a detailed Timeline showing the due date along with actual return dates, and the current Status. Students can locate specific entries using a keyword timeline search bar or instantly filter the list by four distinct status views: All, Active, Returned, and Overdue.

My Attendance Module

This module provides Students with a personal record of their library check-in and check-out sessions. The interface features a structured data table displaying the Date, Check In time, Check Out time, and session Status for each visit. Students can also view their chronological tracking logs through an All-Time History option.

Notification Module

This module alerts students about important system events and updates. Clicking the notification bell in the system header opens a quick drop-down panel showing recent alerts. The dedicated notifications page allows Students to track their library usage.

My Profile Module

This module provides a view-only page where logged-in users can check their personal account details. It displays the user's full name, system ID, official email, contact number, residential address, account status, membership role, and security status.

Limitation

The system requires an active internet connection to synchronize data and operate its cloud-based features.

System capacity and performance are restricted by the database size, serverless execution bounds, and storage limits of the chosen free-tier cloud infrastructure.

Automated fetching of book data via ISBN is often incomplete, requiring librarians to manually calculate and input critical information like Dewey Decimal call numbers and accession numbers.

Authentication is exclusively tied to Microsoft 365 institutional accounts, which prevents access for outside researchers or users without an official school email.

The QR-based automation heavily depends on functional device cameras and readable physical codes, meaning transactions will fail if the hardware is broken or the printed code is damaged.

The searchable catalog focuses strictly on physically available local resources and local thesis soft copies, meaning it does not integrate with enterprise-grade electronic research databases like EBSCO.

The application is constrained by external API rate limits and real-time synchronization quotas, which may restrict the volume of requests processed during peak library hours

REVIEW OF RELATED LITERATURE / SYSTEM

Review of Related Literature

Global Shift Toward Digital Library Transformation

The rapid evolution of digital technologies has significantly impacted how libraries operate, leading to a shift in library services locally and internationally. Despite the widespread acknowledgment of this digital shift, there remains a substantial gap in understanding of how academic libraries are adapting, specifically in integrating digital innovations to enhance user access and engagement. In the study of Distanont et al (2024), there is a critical global trend toward integrating digital innovations to enhance user engagement. This shift is not merely about digitizing records but about strategic adaptation to a landscape where users expect seamless access. This is further supported by Wijaya et al (2025), which emphasizes that libraries relying on manual bookkeeping face inaccurate data recording and time-consuming searches, which hinder the efficiency of library services and reduce the interest of students. As libraries, such as STI College Alabang, manage growing collections and student population, the integration of modern identification and tracking technologies has become essential for maintaining operational integrity and user satisfaction.

Institutional Security and Microsoft 365 Integration

A primary concern in digital transformation is the security of institutional data. Literature on Identity and Access Management (IAM) highlights that existing platforms, like Microsoft O365, provides enterprise-grade security, including Multi-Factor Authentication (MFA). In line with this, Kasahara and Shimayoshi (2022) provided a high-level technical precedent for the use of Microsoft 365 tenants to manage university students. The research highlights the necessity of Multi-Factor Authentication (MFA) and Azure Active Directory to prevent malicious software from compromising student accounts. By implementing M365 authentication, the present study aligns with global best practices that prioritize controls of user access behavior, ensuring that library services are restricted to verified students while offering a self-enrolling convenience that reduces administrative conflict (Kasahara & Shimayoshi, 2022).

QR-Based Scanning and Operational Efficiency The integration of Quick Response (QR) codes represents an opportunity to modernize asset tracking. Paul and Naikar (2024) defined QR codes as powerful tools for instant access to catalog information and operational efficiency. Unlike traditional bookkeeping, QR codes can be scanned using mobile devices or low-cost scanners and QR technology allows for fast data retrieval at a minimal cost (Din & Fazla, 2021). This automation ensures that every transaction is instantly linked to the correct user profile, maintaining a precise digital borrowed history.

However, in practical application, as seen in the study of Mulya and Sari (2024), the transition from conventional systems to QR-web-based applications enables librarians to manage large quantities of data effectively. This technology minimizes manual effort and reduces data-entry errors by automating the link between the identity of the borrower and the library asset (Pebrian & Gunanto; 2025, Wijaya et al., 2025).

Centralized Web-Based Architectures and Real-Time Data Modern Library Management Information Systems (MIS) are increasingly built on robust web frameworks, like JavaScript, Tailwind CSS, TypeScript, or PostgreSQL, to ensure structured and secure development (Rajee et al., 2022, Wijaya et al., 2025). The literature suggests that a module-based approach, which comprises registration and circulation among the university is the most effective way to meet the needs of diverse users (Abdullah et al., 2021). Furthermore, the inclusion of an audit log provides a transparent trail of all system modifications, which suggests that

automated logs are essential for preventing unauthorized changes and ensuring the safety of institutional data.

A benefit identified in the study of Pebrian and Gunanto (2025) is the real-time synchronization of stock quantities. When a student submits a borrowing request using a web interface, the system updates in real-time, providing an up-to-date status that supports data-driven evaluation and strategic decision-making for administrators, such as librarians and even university principals.

Enhancing User Experience through Personalization and Visualization Modern library users expect the same level of convenience found in e-commerce platforms. Beyond simple borrowing, advanced web applications are now incorporating data visualization and personalized recommendations to meet the evolving needs of students (Facunla & Encarnacion, 2024). Guided by Technological Acceptance Theory (TAT), researchers found that tailored web applications can achieve effectiveness rates as high as 97.48% when they focus on usability and reliability. This aligns with the Lumina project goal of providing a personalized student dashboard to help students track their own history and deadlines, thereby increasing their engagement with library resources.

Related Studies and Systems

Local Studies and/or Systems

The research conducted on library management systems highlights the importance of efficient resource tracking and circulation in educational institutions. The studies emphasize the significance of accurate, real-time library inventories to prevent data loss, ensure optimal resource utilization, and seamlessly meet the academic needs of both students and library personnel.

The studies reviewed the functionality of automated library management systems in terms of speed, data handling, accuracy, security, stability, and usability. They compared traditional manual logbook methods with digitized web-based systems, highlighting the massive advantages of integrating QR-code technology and automated workflows in solving administrative challenges, eliminating manual encoding errors, and improving overall library functionality.

Enhanced Information Retrieval: Aemilianum College Inc.'s Library Management System (2023) This research details the development of a web-based Library Management System aimed at overcoming the limitations of physical document deterioration, rising costs, and the frequent misplacement of files. Just as this study ensures seamless access to educational resources through an intuitive and convenient interface, Lumina will provide STI College Alabang with a centralized, reliable digital platform. By transitioning to this web-based solution, Lumina aligns directly with the goal of maximizing the utility of technological resources to make information retrieval, cataloging, and inventory auditing far more efficient (Gito et al., 2023)

Development and Evaluation of an ICT-Based Library Automation System for Northern Iloilo Polytechnic State College Using ISO Standards (2025) This study addresses the inefficiencies of predominantly manual library operations, such as handwritten client logging and inventory updates, by proposing an Information and Communication Technology (ICT)-based Library Automation System. Just as this study demonstrated that transitioning to a QR-integrated digital platform drastically reduces manual workload and improves service delivery, Lumina will implement similar QR-based borrowing and returning transactions for STI College Alabang to ensure high system accuracy, efficiency, and secure user account management (Villoga, 2025).

Library Systems Development: Utilization, Usability and Impact for Private Schools in Negros Occidental, Philippines (2025) This study details the creation and deployment of a web-based Library Management System (LMS) at Colegio de Sta. Ana de Victorias to replace traditional, inefficient manual library processes. This successful implementation of a centralized dashboard and online catalog directly inspires Lumina’s objective to provide STI College Alabang with a searchable online catalog displaying real-time book availability. By adapting these features, Lumina will automate the circulation tracker to reduce processing time

and grant students instantaneous retrieval of information (Lampaza et al., 2025).

Aligning Expectations and Perceptions: Analyzing Library User Satisfaction in Higher Education in Rural Philippines (2025) This research examines the alignment between user expectations and perceptions of academic library services, revealing that modern students hold high expectations for digital service provision and efficient communication channels. Addressing the critical service quality gaps identified in traditional setups, Lumina aims to provide a responsive, user-centered digital framework for STI College Alabang. By digitizing borrowing policies and improving technology availability, Lumina aligns directly with modern students' high expectations for accessible and equitable learning resources (Seno-Sanoria, 2025).

School Management System 1: Library with QR Code Innovation and Book Monitoring: Student Information in Bestlink College of the Philippines (2026) This proposed system introduces a modernized approach to managing school libraries by streamlining the tracking and access of library resources through QR code scanning, an automated fine tracker, and an online book reservation option. Similar to how this system replaces disorganized collections and outdated checking methods, Lumina aims to resolve these exact frustrations at STI College Alabang. By adapting these QR code innovations and online reservation features, Lumina will minimize errors in tracking overdue items, create a much smoother transaction process, and guarantee a highly convenient library experience for both students and librarians (Gargoles et al., 2026)..

School Management System: Library Management System with Online Reservation and Renewal Powered by A.I. Based Adaptive Learning and Skill Development (2026) This study presents a modernized library management system designed to enhance operational efficiency through online book reservation, real-time penalty notifications, and a centralized digital library database. Just as this system significantly automates administrative tasks and improves user engagement, Lumina will implement similar web-based reservation and real-time tracking features. By adapting these digital capabilities, Lumina ensures that STI College Alabang students can seamlessly check book availability and manage their borrowing transactions online, fostering a highly convenient, efficient, and learner-centered academic environment (Mendiola et al., 2026).

Foreign Studies and/or Systems

A Smart QR: Based Library Management System (2024) This study introduces a smart, user-friendly method for managing library materials in tertiary institutions using a QR code system, aiming to replace expensive RFID or IoT alternatives. Lumina adopts this highly cost-effective methodology, utilizing QR codes to empower STI College Alabang students to independently search for books and process checkouts. By scanning unique QR codes, Lumina will drastically reduce the cost, time, and human supervision required for circulation management while digitally managing overdue penalties (Aliyu, Mohammed, & Yakubu, 2024).

A Dual-Validation Mobile Attendance Application for Discipline-Oriented Vocational Schools: Development and Evaluation Using the Waterfall (2025) While focusing on a different educational administrative process, this study highlights the effectiveness of using QR codes to digitize and secure tracking systems that were previously entirely manual and prone to data manipulation. This validates Lumina's core mechanism, demonstrating that by linking unique QR codes to a centralized digital database, Lumina will substantially enhance data integrity, drastically reduce administrative workloads, and provide a highly reliable tool for tracking library resources at STI College Alabang (Kurnia et al., 2025).

Enhancing Library Efficiency With Vmedulife Integrated Library Management Cloud Based System: A Case Study at SVPM’s Institute of Management Malegaon Bk (2025) This study highlights the effectiveness of using a cloud-based Integrated Library Management System (ILMS) to modernize academic institutions by offering a digital catalog, student self-service portals, and real-time analytical reports. Aligning with this cloud-based approach, Lumina will implement a highly secure web architecture to empower students with a self-service dashboard where they can independently search for books, track due dates, and reserve titles. As proven by this study, bringing these operations to the cloud ensures that library services remain continuously accessible while drastically reducing the manual workload of administrators (Wayal, 2025).

Integration of Koha Library Management System with Open-Source Tools for Customized Readers ID Card with QR Codes (2025) This research explores the integration of QR codes into library reader ID cards to ensure quick and error-free library transactions, such as check-ins, check-outs, and gate entries. Validating Lumina's core tracking mechanism, this study demonstrates that replacing traditional manual cards with unique QR codes linked directly to the library's central database substantially minimizes human intervention. By adopting this methodology, Lumina will avoid manual data entry errors, prevent alignment mismatches, and provide a highly reliable, cost-effective tool for tracking resource circulation at STI College Alabang (Koya & Waral, 2025).

Library Automation and Its Impact on Services: A Case Study of the Central Library (2025)

Modern advancements in academic infrastructure emphasize the transition from physical card-based systems to technology-driven procedures that utilize Integrated Library Management Systems (ILMS) to consolidate diverse functions. Research indicates that platforms such as Koha serve as a foundation for contemporary library administration by automating repetitive tasks including acquisitions, circulation, and user account management which effectively reduces the necessity for manual intervention. By replacing traditional manual methods with digital transaction records and automated searching through an Online Public Access Catalog (OPAC), institutions can significantly minimize manual errors and enhance real-time resource tracking. For collegiate environments like STI College Alabang, adopting this automated methodology not only optimizes internal operations and reduces administrative burdens for library personnel but also improves student engagement by providing seamless access to digital and physical resources (Sujana & Putu, 2025).

Synthesis

The global transition from manual library processes to digital platforms is driven by the need to overcome inefficient data recording and time-consuming searches that hinder user engagement. Modern literature establishes that deploying robust, web-based architectures integrated with Quick Response (QR) technology is the most effective strategy for automating resource tracking and minimizing manual data-entry errors. QR codes provide a highly cost-effective alternative to traditional systems, enabling instant catalog access and the seamless synchronization of borrowed assets. Furthermore, centralized cloud-based architectures provide real-time data, allowing administrators to continuously monitor stock quantities and make data-driven decisions while ensuring the safety of institutional records through automated audit logs.

Beyond operational efficiency, recent studies emphasize that automated systems must strictly prioritize institutional security and user-centered design. Implementing enterprise-grade access controls, such as Microsoft 365 and Multi-Factor Authentication (MFA), is crucial for restricting services to verified users and preventing malicious data breaches. Additionally, equipping systems with personalized dashboards, online reservations, and real-time visualizations directly aligns with the modern student's high expectations for convenience, significantly enhancing their overall academic experience.

The proposed project, Lumina, is highly justified as it actively synthesizes the most effective technological advancements identified in both local and foreign literature. By completely replacing outdated, error-prone manual procedures with a centralized web platform, Lumina addresses the critical service quality gaps at STI College Alabang. Its seamless integration of QR-based tracking, secure Microsoft 365 authentication, and real-time inventory auditing ensures a modernized, secure, and highly efficient digital learning environment that effectively meets the evolving demands of both students and library personnel.

Technical Background

Overview of current Technologies to be used in the system

The system is built upon a modern, high-performance tech stack designed for scalability, security, and a premium user experience. At its core, the application utilizes Next.js, a powerful React framework that enables

server-side rendering and optimized routing, ensuring fast load times and seamless navigation. This foundation is reinforced by TypeScript, which provides a robust type system to catch errors early and maintain high code quality across the entire development lifecycle. The combination of these core technologies allows for a highly responsive and maintainable codebase that can adapt to evolving requirements.

For backend infrastructure, the system leverages Supabase, an open-source Backend-as-a-Service that provides a managed PostgreSQL database, integrated authentication, and real-time data synchronization. This allows the system to handle complex data relationships and user security with minimal overhead, utilizing Row-Level Security (RLS) to ensure data integrity at the database level. The backend is further supported by Node.js, providing a scalable runtime environment for executing server-side logic and managing asynchronous operations efficiently.

The user interface is crafted with a focus on aesthetics and accessibility, utilizing Tailwind CSS for rapid, utility-first styling and Radix UI for unstyled, accessible component primitives. Fluid animations and interactive transitions are powered by Framer Motion, while Lucide React provides a consistent and modern iconography set. Data visualization is handled by Recharts, offering clear and actionable insights through responsive charts. Together, these tools create a cohesive design system that feels premium, alive, and intuitive for every user.

Date-fns

Date-fns is a modern JavaScript date utility library that provides a comprehensive and modular set of functions for manipulating and formatting dates. Unlike traditional libraries, it is built with a functional approach, allowing developers to import only the specific functions they need, which significantly reduces the final bundle size. It supports a wide range of locales and provides precise control over date arithmetic, making it indispensable for handling time-sensitive data within the library system.

ESLint

ESLint serves as the primary pluggable and configurable linting tool for the project, used to identify and report on patterns in JavaScript and TypeScript code. By enforcing a consistent coding style and flagging potential bugs or anti-patterns during development, it ensures that the codebase remains clean and maintainable. It integrates directly into the development workflow, providing real-time feedback and maintaining high standards across the entire development team.

Figma

Figma is a browser-based web design and prototyping tool that plays a critical role in the system's UI and UX design phase. It allows for real-time collaboration between designers and developers, ensuring that visual components, layout structures, and user flows are finalized before implementation. By using Figma, the team can create high-fidelity mockups and interactive prototypes that serve as the visual blueprint for the entire web application.

Framer Motion

Framer Motion is a production-ready motion library for React that is used to create fluid, interactive, and high-performance animations. It simplifies the process of adding complex transitions, gestures, and physics-based animations to UI components, making the interface feel more responsive and engaging. In this system, it is utilized to enhance the "premium" feel by providing smooth state transitions and subtle micro-animations that guide user attention.

HTML5-QRCode

HTML5-QRCode is a lightweight and versatile library used for scanning and generating QR codes directly within the web browser. It leverages the device's camera via HTML5 standards to provide a fast and reliable scanning experience for library materials and student identification cards. This technology is a core component of the system's inventory and attendance modules, enabling quick data entry and verification without the need for specialized hardware.

Lucide React

Lucide React is a library of consistent and beautiful icons designed to provide clear visual cues and enhance the overall user interface. Each icon is built as a highly customizable React component, allowing for easy integration with the system's design system and styling tools. These icons are strategically placed throughout the application to improve scannability and ensure that navigation remains intuitive and visually cohesive.

NodeMailer

Nodemailer is to easily and securely send emails from Node.js applications. It eliminates complex configuration by streamlining the email-sending process, allowing developers to handle transactional emails, notifications, and verification codes (OTPs) directly from their backend..

Next.js

Next.js is a comprehensive React framework used for building the full-stack web application, providing essential features like server-side rendering (SSR), static site generation (SSG), and optimized routing. It streamlines the development process by handling complex configurations and offering built-in performance optimizations like image processing and code splitting. This framework is the primary engine of the system, ensuring that the application is both SEO-friendly and exceptionally fast for the end user.

Node.js

Node.js is a cross-platform JavaScript runtime environment that allows the execution of JavaScript code outside of a web browser. In the context of this system, it provides the foundation for the build tools, server-side logic, and various background scripts that handle data processing. Its non-blocking, event-driven architecture makes it ideal for handling the multiple concurrent requests and real-time updates required by a modern library management system.

Nodemailer

Nodemailer is a module for Node.js applications that allows for easy email sending. It is the primary engine used by the system to communicate with the Mailtrap SMTP server, handling the complexities of connection management and message formatting. It supports various transport methods and is utilized here to send professional, HTML-formatted notifications to library members and administrators reliably.

Radix UI

Radix UI is an unstyled, accessible component library that provides the foundational building blocks for creating high-quality design systems. By offering "headless" components that handle complex logic such as focus management, keyboard navigation, and ARIA attributes it allows developers to focus entirely on custom styling. This ensures that the system's interactive elements are fully accessible and standards-compliant while maintaining a unique and premium visual identity.

Recharts

Recharts is a composable charting library built with React components, specifically designed for rendering data-rich analytics and statistics. It uses a declarative approach to define charts, making it easy to build complex visualizations like line graphs, bar charts, and pie charts that are both responsive and interactive. In the system dashboard, Recharts is used to translate raw library data into actionable visual insights for

administrators and librarians.

Sonner

Sonner is an opinionated and lightweight toast component for React that provides customizable, non-intrusive notifications for user actions. It is used to deliver real-time feedback such as success messages after saving data or error alerts when a scan fails without disrupting the user's flow. Its clean design and smooth animations align perfectly with the system's aesthetic, providing a polished way to manage transient application states.

Supabase

Supabase is an open-source Backend-as-a-Service (BaaS) that provides the core infrastructure for the system, including a PostgreSQL database, user authentication, and real-time APIs. It simplifies backend development by offering a unified interface for data management and security, allowing the team to implement complex features like Row-Level Security (RLS) with ease. Supabase serves as the "source of truth" for all system data, ensuring reliability and high availability.

Tailwind CSS

Tailwind CSS is a utility-first CSS framework used for rapidly building and styling custom, responsive user interfaces. Instead of writing traditional CSS, it allows developers to apply styles directly within HTML or JSX using predefined utility classes, which promotes consistency and speed. This framework is essential for maintaining the system's high-density layout and ensuring that the design remains responsive across various device types

TypeScript

TypeScript is a strongly typed programming language that builds on JavaScript by adding static type definitions. It is used throughout the codebase to ensure better code quality, improve developer productivity, and eliminate entire classes of runtime bugs. By providing clear interfaces and type safety for both frontend and backend logic, TypeScript makes the system more robust and easier to refactor as it grows in complexity.

Vercel

Vercel is a cloud platform specifically tailored for Next.js workflows, used for deploying and hosting the web application. It provides a seamless CI/CD pipeline, automatically deploying changes from the repository and offering global edge network delivery for maximum performance. Vercel's infrastructure ensures that the system is always accessible, fast, and secure, with features like automatic SSL and serverless function scaling.

Zod

Zod is a TypeScript-first schema declaration and validation library used for ensuring the integrity of data inputs and API responses. It allows developers to define clear data schemas that are automatically synchronized with TypeScript types, providing both runtime validation and compile-time safety. In this system, Zod is used to validate form submissions, environment variables, and database records, preventing malformed data from causing system errors.

Calendar of Activities

Brainstorming

The proponents spent the initial phase of the development period collaborating and exchanging ideas to identify the common problems in the current library operations of STI College Alabang. This phase focused on planning the core features, such as the QR-based scanning for books, the book reservation workflow, and the inventory management modules for both students and administrators.

Documentation: Chapter I

The developers completed Chapter I with consistent weekly revisions. This chapter serves as the foundation of the study, covering the Project Context, Statement of the Problem, Purpose and Description of the Project, and the Scope and Limitations. It provides a clear roadmap for how the Web-Based Library Book Reservation, Borrowing, and Inventory Management System will function.

Documentation: Chapter II

Chapter II was developed and refined to provide theoretical support for the project. This section includes the Review of Related Literature and Related Systems, comparing existing library software with the proposed QR-based system to ensure the project offers unique and helpful features for the school.

Documentation: Chapter III

The developers completed Chapter III to outline the technical side of the project. This chapter explains the software and hardware requirements, the calendar of activities, and the specific technologies used, such as Microsoft O365 for user authentication, providing a clear overview of the development timeline.

Diagrams

The developers designed the Entity Relationship Diagram (ERD) and Use Case Diagram for the system. These diagrams were revised multiple times to accurately represent how the database stores book information and how different users such as students and admins interact with the library system’s features

Flowchart

The developers created detailed flowcharts to illustrate the step-by-step processes of the system. These charts show the workflow for scanning QR codes, the logic for borrowing and returning books, and how notifications are sent to students when their reservations are approved.

GanttChart

MONTH

FEBRUARY

MARCH

APRIL

MAY

ACTIVITIES

Requirements Gathering

Title Proposal

Brainstorming

Researching

Documentation

Chapter 1: Introduction

Chapter 2: Review Related Literature

Chapter 3: Technical Background

System Development

Transcript of Interview

Resources

Specification:

Hardware Requirements

Processor

Intel Core i5-12400

Graphics Card

AMD Radeon RX 580

Memory

32 GB

Disk Storage

500 GB

Table 1 Hardware Requirements

Software Requirements

Operating System

Windows 11

Integrated Development Environment (IDE) and/or Code Editors

Visual Studio Code

Database

Supabase

Programming Languages and other Technologies

Next.js, TypeScript,

Documentation

Microsoft 0365 Word

Table 2 Software Requirements

Requirements Analysis

Minimum Hardware Requirements (For Development)

MINIMUM

RECOMMENDED

Processor

Intel Core i3-10100 @ 3.6- GHz (4 cores, 8 threads)/ AMD Ryzen 3 3200G @ 3.6GHz (4 cores, threads)

Intel Core i5-11400 @ 2.60GHz (6 cores, 12 threads) / AMD Ryzen 5 5600X

Memory

8GB DDR4

3.7GHz (6 cores, 12 threads) 16GB DDR4

Graphics Card

Integrated Graphics (e.g., Intel UHD Graphics 630 / AMD Radeon Vega 8)

Dedicated Graphics (e.g., NVIDIA GeForce GTX 1660 6GB / AMD Radeon RX 5600 XT 6GB)

Disk Storage

128GB

256GB or more

Table 3 Minimum Hardware Requirements (For Development)

Minimum Hardware Requirements (For Users)

Processor

Intel Core 2 Duo E8400 @ 3.0GHz / AMD Athlon II X2 250 @3.0GHz.

Memory

4GB RAM

Graphics

Integrated sound card (100% DirectX 10 compatible)

Table 4 Minimum Hardware Requirements (For Users)

METHODOLOGY

This project uses the Iterative Development method to build and set up A Web-Based Library Book Reservation, Borrowing, and Inventory Management System with QR-Based Scanning for STI College Alabang. We chose this approach for one main reason: it allows us to build the system and write our research paper at the same time. Instead of waiting to finish all the paperwork before writing code, this method lets us

work on small parts of both tasks together. This helps us make steady progress and ensures our written report always matches the real, working system.

[image omitted]

Stage 1 Planning

In this first step, we plan the main features we want to create for a specific part of the library system, such as the login page or the QR scanner. At the same time, we write down these initial plans and requirements directly into our project document.

Stage 2 Design

Next, we design how that specific part of the system will look and work. We map out the user screens and database layouts for that feature. As soon as the design is ready, we add these sketches and diagrams to our paper.

Stage 3 Implementation

This is the step where we actually write the code and build the feature we just designed, like creating the book inventory table. Working this way allows us to draft the technical descriptions in our manuscript while the coding is fresh in our minds.

Stage 4 Testing

Once the feature is coded, we test it carefully to make sure there are no errors. We check that the QR codes scan properly and that books can be borrowed without issues. All the test results and fixes are written straight into our report.

Stage 5 Evaluation

In this final step of the cycle, we look at the tested feature to see what works and what needs to be fixed. Once this part is reviewed and finalized, we update our paper to match it perfectly, and then we start the cycle all over again for the next system feature.

[image omitted]

Figure 1 Entity Relationship Diagram 1

[image omitted]

Figure 1.1 Entity Relationship Diagram 2

[image omitted]

Figure 2 Data Flow Diagram level

Figure 2.1 Data flow diagram level 1

[image omitted]

Figure 3 Entry Authentication flowchart

[image omitted]

Figure 4 Super-Admin Dashboard Flowchart

[image omitted]

Figure 4.1 Super-Admin Inventory Flowchart

[image omitted]

Figure 4.2 Super-Admin Circulation Desk Flowchart

[image omitted]

Figure 4.3 Super-Admin Borrowing Logs Flowchart

[image omitted]

Figure 4.4 Super-Admin Analytics Flowchart

[image omitted]

Figure 4.5 Super-Admin User Directory Flowchart

[image omitted]

Figure 4.6 Super-Admin Attendance Logs Flowchart

[image omitted]

Figure 4.7 Super-Admin Setting & Policies Flowchart

[image omitted]

Figure 4.8 Super-Admin Audit Logs Flowchart

[image omitted]

Figure 4.9 Super-Admin Profile Flowchart

[image omitted]

Figure 5 Librarian Dashboard Flowchart

[image omitted]

Figure 5.1 Librarian Inventory Flowchart

[image omitted]

Figure 5.2 Librarian Circulation Desk Flowchart

[image omitted]

Figure 5.3 Librarian Borrowing Logs Flowchart

[image omitted]

Figure 5.4 Librarian Analytics Flowchart

[image omitted]

Figure 5.5 Librarian User Directory Flowchart

[image omitted]

Figure 5.6 Librarian Attendance Logs Flowchart

[image omitted]

Figure 5.7 Librarian Settings & Policies Flowchart

[image omitted]

Figure 5.8 Librarian Profile Flowchart

[image omitted]

Figure 6 Student-Assistant Dashboard Flowchart

[image omitted]

Figure 6.1 Student-Assistant Inventory Flowchart

[image omitted]

Figure 6.2 Student-Assistant Circulation Desk Flowchart

[image omitted]

Figure 6.3 Student-Assistant My Attendance Flowchart

[image omitted]

Figure 6.4 Student-Assistant Attendance Logs Flowchart

[image omitted]

Figure 6.5 Student-Assistant My Borrow Flowchart

[image omitted]

Figure 6.6 Student-Assistant Borrowing Logs

[image omitted]

Figure 6.7 Student-Assistant Profile Flowchart

[image omitted]

Figure 7 Student Dashboard Flowchart

[image omitted]

Figure 7.1 Student Catalog Flowchart

[image omitted]

Figure 7.2 Student My Borrow Flowchart

[image omitted]

Figure 7.3 Student My Attendance Flowchart

[image omitted]

Figure 7.4 Student Profile Flowchart

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APPENDIX A. RESOURCE PERSONS

RESOURCE PERSON

The developers of this capstone project would like to express their gratitude to several individuals whose contributions were invaluable to its successful completion. We are particularly indebted to our esteemed capstone advisor, Mr. John Renaund L. Baybay. His insightful guidance, unwavering support, and vast knowledge in the field were instrumental in shaping this project's direction. Mr.Baybay's extensive knowledge and wealth of experience have left an indelible mark on the academic research and daily pursuit of the developers.

Furthermore, the authors express their sincere gratitude to the faculty at STI College Alabang. Their unwavering support, patience, and insightful feedback throughout the project journey significantly enhanced the outcome of the Capstone Project.

APPENDIX B. PERSONAL TECHNICAL VITAE

Curriculum Vitae of

Dadea, Jerico B.
300 Marcelo 2 Bagumbayan, Taguig City
Jericodadea26@gmail.com
09854036633

EDUCATIONAL BACKGROUND

Level	Inclusive Dates	Name of school/ Institution
Tertiary	2023 - Present	STI College Alabang
Vocational/Technical	2018 - 2020	Amtic National High School
High School	2014 - 2018	Amtic National High School
Elementary	2008 - 2014	Basag Elementary School

PROFESSIONAL OR VOLUNTEER EXPERIENCE

Inclusive Dates	Nature of Experience/ Job Title	Name and Address of Company or Organization
2024	NSTP	STI College Alabang

AFFILIATIONS

Inclusive Dates

Name of Organization

Position

2024 - Present

2024 - Present

Computer Society

Coders Club

Technical

Technical

.

SKILLS

SKILLS

Level of Competency

Date Acquired

SAP

Beginner

2025

SAP

Advance

2025

JAVA

Beginner

2023

TRAININGS, SEMINARS, OR WORKSHOPS ATTENDED

Inclusive Dates

Title of Training, Seminar, or Workshop

2024

2023

2023

Key IT Trends to Watch

Step Info Success Pre Employment Essentials for the future

What’s Next In Tech

Curriculum Vitae of

Floriano, Kayle Andrew P.

Blk 5 Lot 46 Cristina Homes San Pedro Laguna

kaylefloriano1826@gmail.com

09615091963

EDUCATIONAL BACKGROUND

Level

Inclusive Dates

Name of school/ Institution

Tertiary

2023 - Current

STI College Alabang

Vocational/Technical

2021 - 2023

Laguna Northwestern College

High School

2017-2021

Muntinlupa National High School

Elementary

2011-2017

Mountaintop Christian Academy

PROFESSIONAL OR VOLUNTEER EXPERIENCE

Inclusive Dates

Nature of Experience/

Job Title

Name and Address of Company or Organization

2023

NSTP

STI College Alabang

AFFILIATIONS

Inclusive Dates

Name of Organization

Position

N/A

Listed in reverse chronological order (most recent first).

SKILLS

SKILLS

Level of Competency

Date Acquired

SAP

Beginner

2025

CSS

Beginner

2024

HTML

Beginner

2024

JAVA

Beginner

2023

TRAININGS, SEMINARS, OR WORKSHOPS ATTENDED

Inclusive Dates

Title of Training, Seminar, or Workshop

2024

Cyber Awareness Seminar

2023

Elms Tech Tool

Curriculum Vitae of

Godwyn Neri
12A Laos, Better Living Paranaque
kuyagodwyn@gmai.com
09935223815

EDUCATIONAL BACKGROUND

Level
Inclusive Dates
Name of school/ Institution

Tertiary
2023 - Present
STI College Alabang

Vocational/Technical
2021 - 2023
Siniloan Integrated National High School

High School
2017 - 2021
Gov. Felicisimo T. San Luis National Agro Industrial Integrated High School

Elementary
2011 - 2017
Kapatalan Elementary School

PROFESSIONAL OR VOLUNTEER EXPERIENCE

Inclusive Dates
Nature of Experience/
Job Title
Name and Address of Company or Organization

2023
NSTP

STI College Alabang

AFFILIATIONS

Inclusive Dates

Name of Organization

Position

2025 - Present

Coders Club

Auditor

SKILLS

SKILLS

Level of Competency

Date Acquired

ESP32 & IoT Development

Intermediate

2025

TRAININGS, SEMINARS, OR WORKSHOPS ATTENDED

Inclusive Dates

Title of Training, Seminar, or Workshop

2024

2023

2023

Key IT Trends to Watch

Step Info Success Pre Employment Essentials for the future

What’s Next In Tech

Curriculum Vitae of

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0945-972-7869/0949-838-1414

EDUCATIONAL BACKGROUND

Level
Inclusive Dates
Name of school/ Institution
Tertiary
2023 - Present
STI College Alabang
Vocational/Technical
2021 - 2023
Tunasan National High-School
High School
2017 - 2021
Poblacion National High-School
Elementary
2011 - 2017
Poblacion Elementary School

PROFESSIONAL OR VOLUNTEER EXPERIENCE

Inclusive Dates
Nature of Experience/
Job Title
Name and Address of Company or Organization
2023
NSTP
STI College Alabang

Listed in reverse chronological order (most recent first).

AFFILIATIONS

Inclusive Dates

Name of Organization

Position

2024 - Present

Computer Society

Technical Officer

Listed in reverse chronological order (most recent first).

SKILLS

SKILLS

Level of Competency

Date Acquired

JAVA

Beginner

2023

JAVA

Beginner

2023

TRAININGS, SEMINARS, OR WORKSHOPS ATTENDED

Inclusive Dates

Title of Training, Seminar, or Workshop

2023

Basic Training

2023

MARPOL Training

Curriculum Vitae of

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#21A San Nicolas de Tolentino st.

Purok 2 Brgy. Cupang, Muntinlupa City

kennethjaemstolentino4@gmail.com
0969.166.0480

EDUCATIONAL BACKGROUND

Level
Inclusive Dates
Name of school/ Institution

Tertiary
2023-Present
STI College Alabang

Vocational/Technical
2021-2023
Cupang Senior High School

High School
2017-2021
Muntinlupa Business High School Main

Elementary
2010 - 2017
Cupang Elementary School Main

PROFESSIONAL OR VOLUNTEER EXPERIENCE

Inclusive Dates
Nature of Experience/
Job Title
Name and Address of Company or Organization

2023
NSTP
STI College Alabang

AFFILIATIONS

Inclusive Dates
Name of Organization
Position

2024

Computer Society

Technical Officer

SKILLS

SKILLS

Level of Competency

Date Acquired

SAP

Beginner

2025

SAP

Advance

2025

HTML

Beginner

2024

JAVA

Beginner

2023

TRAININGS, SEMINARS, OR WORKSHOPS ATTENDED

Inclusive Dates

Title of Training, Seminar, or Workshop

2024

Cyber Awareness Seminar

2023

Elms Tech Tool

Transcript of Interview

Interviewer: Today, we are here for an interview about our research title: "A Web-Based Library Book Reservation, Borrowing, And Inventory Management System With Qr-Based Scanning For STI College Alabang." Are there any features you can suggest we add?

Interviewee: Do you have a set of questions prepared? First, is the design origin intended to be online or offline?

Interviewer: What we have in mind is to make it both online and offline. It can function offline so that when there's no internet, the system will still work. But when there is internet connection, it will sync to the cloud, similar to the concept of Microsoft 365. We designed it this way so that when recording data, since books have ISBN barcodes, we can simply scan them, and the title and author will automatically be fetched.

Interviewee: The first group I talked to had a Windows-based system, but your concept can indeed be a web-based system that works both online and offline. Okay, go ahead. What are your questions regarding the features?

Interviewer: We just want to ask if there are any specific features you'd like us to add? Also, our system is capable of generating...

Interviewee: My first question is, what are the basic requirements of a library system? Do you have an idea?

Interviewer: Borrowing, sir. We will allow that in the system.

Interviewee: Usually, it requires more than just borrowing. What modules make up the basic system of a library?

Interviewer: It will record the data of the books. But it's not just about recording.

Interviewee: Let's ask AI: "What are the basic modules or requirements of a library system based on standards used widely in universities abroad and here in the Philippines?" Also, we need to clarify is this library system intended for Senior High School (SHS) or the college level?

Interviewer: Actually, sir, it's intended for the whole school.

Interviewee: That's possible. There is a specific system structure used for Senior High, but we can confirm the appropriate standards with your school librarian.

Interviewer: I think a unified system is possible because we plan to use the students' school Microsoft accounts for the logins.

Interviewee: Usually, college and senior high school libraries are managed separately. When you go to a library, books are categorized by specific numbers—you don't just search for them randomly. They follow a classification system, like the Dewey Decimal System, where specific numbers correspond to specific subjects. Aside from the ISBN, there is a traditional coding system you must follow. An integrated library system, like Koha or Follett Destiny, can actually manage both levels. They handle this by using the same software but assigning different branch codes, allowing the librarian to see which book belongs to the high school wing and which to the college wing. You need to consider these library standards and perhaps look into what top universities use, such as enterprise-grade library platforms.

Interviewer: We also have an idea regarding the data. Since the books have ISBN codes, we can use those codes to automatically fetch all the information about the book from the internet and store it locally. We will scan the books one by one, place the retrieved data into the local database, and then sync it to the cloud.

Interviewee: Take note that libraries use specific classification numbers. The ISBN is good for finding a unique title, but the Dewey Decimal numbers are what librarians use to instantly identify a book's category. What will be your system's process flow? You need to follow standard library processes, otherwise, no matter how good the system you build is, it will be incorrect.

Interviewer: So we should integrate the standard library process—for example, the specific classification codes used for Filipiniana books—because we can't just rely on ISBNs and titles. In our school's library, the numbering system seems mixed up and not consecutive, but there is an underlying structure recorded in an

Excel file.

Interviewee: Even if they seem mixed, books with similar classification codes are grouped together on the shelves to make them easy to locate. That's why your actual client here must be the librarian. You should research Koha, an open-source library system. You need to benchmark against it because your panelists might ask you during defense: "Why should we use your system when Koha is free and already available?" You need to figure out what makes your system better than existing open-source platforms.

Interviewee: Koha is used globally and strictly follows international standards. Before you consult the librarian, you should familiarize yourself with library technicalities. For example, there's the OPAC (Online Public Access Catalog)—this is where students search for books. There's also Cataloging, which handles the processing of book data. Then there's Circulation, which manages loans, returns, transfers, and acquisitions. You will also need a Reporting module.

Interviewer: We already planned a feature for that, sir. If the age limit of a book in the library is, for instance, 20 years, and it was published in 2007, it should be pulled out next year. Our system will have a feature to notify the librarian for the disposal of outdated books.

Interviewee: Yes, you can include predictive reporting for book disposal. To improve your system's concept, I highly recommend visiting the National Library to observe their system. Another thing our current library lacks is a local repository for thesis papers. Would you like to include soft copies (PDFs) of local research and theses in your system?

Interviewer: We can put the thesis files there, sir.

Interviewee: Students are required to submit their theses to the library. If you host soft copies locally, the system can establish that those studies are the intellectual property of the creators and the school.

Interviewer: How will the system be useful if it is strictly designed for the librarian and not the students? We originally planned the system purely for borrowing and reservation.

Interviewee: There is no library system made solely for the librarian! To make your system unique, you could add an offline access feature to view soft copy resources hosted on a local server. This allows students to access research papers directly from the library system's database without triggering major copyright violations on the public internet.

Interviewer: We would need to place computers in the library for that. Are the current ones usable?

Interviewee: Yes, they are. Libraries are actually required to have computer stations for students to access digital resources.

Interviewee: We will probably add a mapping feature to our system. It would function like Google Maps to show the exact physical location of the books inside the library.

Interviewer: Yes, when a book is searched, the details and its specific shelf location should be displayed. To make your user interface and OPAC even more unique, you could integrate connections to scholarly sites like Google Scholar or Research Hub for when the system is online. I highly suggest you benchmark by visiting the National Library; it's free. You can also visit the Muntinlupa Public Library to compare systems, but use the National Library to establish your core standard.

Interviewee: Before you start coding, you need to finalize the logic design and system flow. Many developers make the mistake of coding the system first and thinking about the logic later. Use the Agile methodology and map out a flowchart first to see how modules connect. An integrated library system means all tasks must sync. If a book is checked out in the circulation module, it must automatically update in the OPAC so students won't see it as available. Cataloging is the brain that processes data, OPAC is the face where students log in to reserve books, and Circulation handles limits, returns, and inventory.

Interviewer: We have included inventory management, sir.

Interviewee: Good. Ensure you add the baseline features that are missing before focusing on unique additions. Your system must cover these standard library modules first.

Interviewer: We are working on a library book reservation, borrowing, and inventory management system with code-based scanning. Our plan is to provide both online and offline functionality, where the system works without internet but syncs to the cloud once connected. For data recording, we intend to use ISBN barcodes to automatically retrieve book titles and authors from the internet.

Interviewee: This should be a web-based online/offline system. Beyond basic recording, you must determine the requirements based on international and local university standards. For instance, a library requires a coding system like the Dewey Decimal System; you cannot categorize books by ISBN and title alone.

Interviewer: Regarding basic requirements, I believe the system needs to handle borrowing and recording book data. I also think we can integrate the school's Microsoft login for user access.

Interviewee: Using a unified system for both Senior High School (SHS) and college is recommended for library literacy, as it teaches students how to navigate university libraries early on. An Integrated Library System (ILS) like Koha or Follett Destiny can manage both levels, using branch codes if the collections are in separate buildings, The Dewey Decimal System remains the international standard you should use.

Interviewee: Large institutions use enterprise-grade Library Service Platforms (LSP) and Electronic Resource Management (ERM). While they use online resources like EBSCO, your system will focus on local resources—those physically available in the library.

Interviewer: My idea for the ISBN codes is to pull all book information from the internet and store it locally in our database.

Interviewee: The real challenge is the process flow. Even with QR codes and borrowing features, the system is incorrect if it doesn't follow standard library flows. For the tertiary level, you must ensure the system has a Z39.50 feature to allow for copy cataloging from larger libraries. You must research these proper processes and standard codes because a professional librarian will require them.

Interviewer: At the National Library, we saw they use barcodes for attendance (time in/out) and borrowing. They have a website for searching called OPAC, which sometimes includes logos that direct you to a borrowing page. For physical books, we reserve them using an account and a library ID card. We don't have access to the shelves; the staff retrieves the books for us at the desk.

Interviewee: That concept is likely because they are protecting a large collection of books. In that system, the circulation area is where you return the books, and they are not allowed outside the library. They use QR codes to verify returns at the guard station to ensure no books are taken out without being cleared.

Interviewer: We want to adapt parts of that. Our plan is to use a QR code scanner where students have digital library cards on their phones, which they can also print.

Interviewee: You need to be careful with the technicality of library standards. Even if a web app has the default UI (like minimize/maximize/exit), the process flow must meet minimum librarian requirements. You mentioned not wanting to use a standard system like Dewey because our library is small, but your system must be future proof. If the library grows beyond 5,000 books, a limited system will fail.

Interviewer: Our system will include reservation functionality, though it's optional. To borrow, a student brings a book to the librarian, who scans the student’s QR library card and then the book’s QR code to register the transaction. We also want to set borrowing limits for example, one day for textbooks and longer for fiction.

Interviewee: Regarding user roles, you should have Admin, Librarian, and Staff/Student Assistant accounts. However, inventory and resource management must be restricted to those with high accountability

(Librarians/Admins). Student assistants should only handle basic tasks like borrowing and returning, as inventory involves resources that cost money.

Interviewer: For the inventory, we added a feature were scanning an ISBN automatically retrieves the book title and information for the OPAC/catalog.

Interviewee: Check what standards are required for cataloging. Usually, it includes title, edition, ISBN, subject headings, keywords, call number, and location status. I suggest you consult a professional librarian, like Miss Bet, or a Library Science graduate to ensure the process flow is correct.

Interviewer: We also want to make it innovative. We are thinking of sorting books by category using machine learning or analytics. We are also considering adding a login/logout feature for student attendance.

Interviewee: To maximize the system for students, make it a "Study Body" tool. Instead of just borrowing, allow them to use it for research, taking notes, or creating reviewers with linked resources and PDFs. You could also include a local server for digital theses and offline e-books to avoid storage and copyright issues.

Interviewer: For the payment of fines or violations, we are thinking of a walk-in process where the library manually inputs the payment into the system.

Interviewee: That’s fine. The library can handle the manual input. Regarding penalties, if a student has an overdue book, the system should calculate a penalty—for example, five pesos per day. However, the system must be configurable so the librarian can decide the daily rate or even "waive" the penalty for valid reasons, such as a death in the family.

Interviewer: We are planning to visit other schools and libraries to conduct further research on their processes.

Interviewee: You should search for libraries that follow international standards. Public libraries may not always be organized, but the National Library is a good benchmark. I recommend looking at "Big Four" universities like UST, La Salle, or Ateneo, as they invest heavily in their library systems. You need to observe their traditional library processes, even if they still use paper for some tasks.

Interviewer: We were looking into OPAC (Online Public Access Catalog). In our system, it’s mostly just the catalog without a separate front-end search engine.

Interviewee: OPAC is essentially the front-end search engine where the bibliographic details author, date of publication, and summaries live. Your system needs an inventory entry module for this bibliographic data, as well as resource management and authority control. You also need a User Management or CRM module to track who can borrow and their transaction history.

Interviewer: For user access, we plan to restrict logins to those with a Microsoft school account. We will filter registration so only emails with the school's domain (e.g., @alabang.sti.edu.ph) can enter the system.

Interviewee: Ensure you include the faculty and staff as well, as their emails might have slightly different indicators (like.edu.ph or specific staff codes). If the filter identifies them as being from the Alabang branch, it will work. For your continued research, focus on schools with the budget to properly maintain these systems, as they will provide the best examples of high-standard library management.

Interviewer: In our previous interviews, we mentioned that the system would have both online and offline features. However, because we have integrated cloud-based features that require continuous internet access, we are no longer able to include an offline mode. This shift occurred because the system transitioned to a cloud-based framework.

Interviewee: Since it is cloud-based, that inherently means it is online. What is the core concept of your system now? Can they access it from home?.

Interviewer: Yes, sir. Users can access the system remotely from their homes because it is now a web-based platform. Even when they are physically at the school library, the system remains accessible. Students can use the online catalog to check for available book copies and make reservations.

Interviewee: The inventory should definitely be reflected there. But what happens if there is no internet connection? Will students still be able to borrow books?.

Interviewer: If there is no internet, the website will be unavailable, but the library itself will not close. During such instances, the library will revert to manual features and manual transactions. Once the internet connection is restored, the system will return to normal operation. Our system's technical limitations specifically state that it requires continuous internet access to function.

Interviewee: Why did you decide to focus primarily on the online aspect?.

Interviewer: We decided to focus on the online system because the institution (STI) provides consistent internet access, so we prioritized optimizing those features.

Interviewee: How does the process work when a student enters the library?.

Interviewer: Students will use their library cards for attendance. Even though the system is online and they can log in from anywhere, physical entry to the library still requires a screening process. We have integrated an attendance feature where the system scans the student's library card for time-in and time-out purposes.

Interviewee: So, the library card is the basis for entry?.

Interviewer: Yes, sir. Students will use a QR code on their cards for both entry and exit tracking. This same QR code is also used for the book borrowing process. To summarize, the system requires internet access to operate these features; otherwise, the library must utilize manual transactions during a power outage or internet downtime.

INTERVIEWERS:

DADEA, JERICO B.

FLORIANO, KAYLE ANDREW P.

NERI, GODWYN O.

RESSURRECION, RHED R.

TOLENTINO, KENNETH JAMES B.

INTERVIEWEE:

Mr. Ricardo M. Ricson Academic Head, STI College Alabang

Interviewer: We are planning to transition the library's manual transaction process into a digital one. Instead of manual writing, we want to use ISBN codes and student ID numbers for a barcode-based system.

Interviewee: Before you transition to digital, all books must be classified with a call number. The system must be able to identify if a book is copyrighted within the last five years, which is the standard for professional and updated collections, unless the book is historical. Subject classification typically ranges from 000 to 999.

Interviewer: What specific classification system should we be using?

Interviewee: You should use the Dewey Decimal Classification (DDC) system, as it is the standard unless you are using the Library of Congress system. For your research, I recommend studying existing systems like Platonics or Sophia. These will give you a clear idea of how to handle charging and discharging (borrowing/returning), inventory, and loan records.

Interviewer: Our current system uses generated QR codes for every book and scans student library cards for identification.

Interviewee: That works, but the system must also reflect the library's rules and regulations, including policies on loan durations and penalties. You also need to manage different user types, such as Senior High School and College students, faculty, and even outside researchers who might visit with a referral letter.

Interviewer: When we add a new book to the system, what information is essential besides the title and author?

Interviewee: A book should not be cataloged or assigned a call number until it is fully paid for. Once it is, you must assign an accession number. While many books may share the same copyright date (e.g., 2020), the accession number is unique to every single copy in the library. This allows you to track exactly which copy is missing if one is lost.

Interviewer: We are currently using ISBN codes to automatically "fetch" data like the title and cover photo from the internet.

Interviewee: Automated data fetching is helpful, but it is often incomplete. The most critical inputs that a librarian must verify or input manually are the call number (classification) and the accession number. The call number is derived from the subject and the author’s surname to ensure the book is placed correctly in the collection.

INTERVIEWERS:

DADEA, JERICO B.

FLORIANO, KAYLE ANDREW P.

NERI, GODWYN O.

RESSURRECION, RHED R.

TOLENTINO, KENNETH JAMES B.

INTERVIEWEE:

Ms. Elizabeth A. Bumalay Faculty, STI College Alabang

TAB 2: Requirement Documentation

Requirements Documentation

This specific part defines the software features of the web-based system of Lumina: A Web-Based Library Book Reservation, Borrowing, and Inventory Management System with QR-Based Scanning for STI College Alabang. The requirements stated below are formulated based on the data gathered, standard operating procedure analysis, and key stakeholder consultations conducted with the library staff and administration of STI College Alabang, as well as particularly selected information based on the specific objectives of this study. The defined requirements below are expected to cater to the operational needs of STI College Alabang to enhance library transaction efficiency, inventory tracking accuracy, and asset security. The requirements below also describe how all the authorized users Super Administrators, Librarians, Student Assistants, and Students of STI College Alabang can interact with the system.

4.1. Functional Requirements

4.1.1. Log In (Super Admin)

REQ001. The super admin shall be able to log in securely using Microsoft O365 institutional account credentials.

REQ002. The system shall restrict access to administrative tools to ensure that only authorized super administrative personnel can manage backend data.

4.1.2. Dashboard Module (Super Admin)

REQ003 . The super admin shall be able to view a real-time overview displaying metric summary counters for daily visitor traffic, registered users, total book inventory, and active loans.

REQ004. The super admin shall be able to utilize quick-access navigation shortcuts to navigate directly to the Circulation Desk, Book Inventory, User Directory, Visitor Logs, Audit Logs, and Analytics modules.

REQ005. The super admin shall be able to view a visual gallery highlighting the most recently cataloged library assets on the main dashboard interface.

4.1.3. Inventory (Super Admin)

REQ006. The super admin shall be able to automatically fetch book metadata (title, author, cover art) via automated ISBN lookup.

REQ007. The super admin shall be able to manually finalize catalog records by entering Dewey Decimal numbers, book categories, total copy counts, and physical shelf locations.

REQ008. The super admin shall be able to utilize a Batch Catalog Operations panel equipped with a drag-and-drop Import Sheet tool for uploading inventory spreadsheets.

REQ009. The super admin shall be able to utilize an ISBN Fallback text field for bulk sequential lookups and automatic cover art retrieval.

REQ010. The super admin shall be able to export active catalog data with custom column selections into Spreadsheet, CSV, JSON Data, or Covers ZIP file formats.

4.1.4. Circulation Desk (Super Admin)

REQ011. The super admin shall be able to scan student ID QR codes and physical book QR codes using camera or barcode input devices.

REQ012. The system shall automatically link the borrower to the scanned book item during circulation transactions to eliminate manual data entry.

4.1.5. Borrowing Logs (Super Admin)

REQ013. The super admin shall be able to view a centralized, view-only list tracking all historical and active borrowing and returning transactions.

REQ014. The super admin shall be able to view complete transaction details including borrower name, borrower ID, book title, author, borrow date, due date timeline, and return date.

REQ015. The super admin shall be able to identify transaction statuses through color-coded status tags (Active, Returned, Overdue).

REQ016. The super admin shall be able to search borrowing records by keyword and filter logs using status tab buttons (All, Active, Returned, Overdue).

4.1.6. Analytics (Super Admin)

REQ017. The super admin shall be able to view system performance metrics filtered by custom timeframes (7 days, 30 days, or 1 year).

REQ018. The super admin shall be able to view summary counters for daily visitor traffic, active book borrows, total inventory volumes, and registered library members.

REQ019. The super admin shall be able to view line graphs representing daily traffic distributions and circulation trends over time.

REQ020. The super admin shall be able to view a donut chart breaking down asset allocation across returned, active, and overdue statuses.

REQ021. The super admin shall be able to view a demand leaderboard list showcasing the highest-demand catalog books based on borrow counts.

4.1.7. User Directory (Super Admin)

REQ022. The super admin shall be able to view a master list of all registered accounts displaying user full names, emails, unique IDs, system roles, account statuses, and academic programs.

REQ023. The super admin shall be able to search user accounts by name or email address and narrow lists using filter buttons.

REQ024. The super admin shall be able to click on any user entry to open their detailed profile page and inspect contact information.

REQ025. The super admin shall be able to update user account settings, including changing system roles, modifying account statuses, and archiving user profiles.

4.1.8. Attendance Logs (Super Admin)

REQ026. The super admin shall be able to track student traffic in real time by scanning student library cards at entry and exit points.

REQ027. The super admin shall be able to view a list of checked-in and checked-out students along with exact entry and exit timestamps.

4.1.9. Setting and Policies (Super Admin)

REQ028. The super admin shall be able to define maximum circulation limits and reservation limits.

REQ029. The super admin shall be able to configure student academic programs and organize catalog book categories.

REQ030. The super admin shall be able to create and broadcast system-wide announcements to all users.

REQ031. The super admin shall be able to access a Q&A support log to assist with user troubleshooting.

REQ032. The super admin shall be able to execute backend maintenance tools including batch account resets, automated email dispatches, and configuration testing.

4.1.10. Audit Logs (Super Admin)

REQ033. The super admin shall be able to view a security table tracking action timestamps, administrator/user identities, action types, affected entities, and change descriptions.

REQ034. The super admin shall be able to click a dedicated button on log entries to view granular technical differences (diffs).

REQ035. The super admin shall be able to search audit logs by name, action type, or reason, sort logs via entity/action dropdown filters, and execute live data refreshes.

REQ036. The super admin shall be able to import and export audit log records in Spreadsheet CSV and developer JSON formats.

4.1.11. Notification (Super Admin)

REQ037. The super admin shall be able to click a header notification bell icon to view a drop-down panel displaying recent system alerts.

REQ038. The super admin shall be able to access a dedicated notifications page equipped with quick filter tabs and a search bar to manage alerts.

4.1.12. My Profile (Super Admin)

REQ039. The super admin shall be able to view a read-only profile page displaying their full name, system ID, official email, contact number, residential address, account status, membership role, and security status.

4.1.13. Log In (Librarian)

REQ040. The librarian shall be able to log in securely using Microsoft O365 institutional account credentials.

4.1.14. Dashboard (Librarian)

REQ041. The librarian shall be able to view real-time metrics for daily visitor counts, registered users, total book inventory, and active loans.

REQ042. The librarian shall be able to use quick navigation shortcuts to access the User Directory, Audit Logs, and Analytics, as well as review recently cataloged assets.

4.1.15. Inventory (Librarian)

REQ043. The librarian shall be able to manage book collections using ISBN automated lookups, manual cataloging fields (Dewey Decimal, shelf locations, copy counts), spreadsheet import tools, fallback lookups, and customizable catalog exports.

4.1.16. Circulation Desk (Librarian)

REQ044. The librarian shall be able to issue and return books by scanning student ID QR codes and book QR codes.

4.1.17. Borrowing Logs (Librarian)

REQ045. The librarian shall be able to access a view-only list of borrowing records with search tools, status filters (All, Active, Returned, Overdue), and timeline details.

4.1.18. Analytics (Librarian)

REQ046. The librarian shall be able to analyze time-filtered reports (7 days, 30 days, 1 year) including visitor trends, active borrow counts, status donut charts, and book demand leaderboards.

4.1.19. User Directory (Librarian)

REQ047. The librarian shall be able to view master user lists, search user accounts, open profile views, and update student account statuses specifically between Active and Pending.

4.1.20. Attendance Logs (Librarian)

REQ048. The librarian shall be able to log student entry and exit timestamps in real time by scanning student library cards

4.1.21. Setting and Policies (Librarian)

REQ049. The librarian shall be able to manage circulation/reservation limits, academic programs, book categories, post system announcements, and review Q&A support logs.

4.1.22. Notification (Librarian)

REQ050. The librarian shall be able to monitor system alerts via header bell drop-down panels and search/filter notifications on a dedicated page.

4.1.23. My Profile (Librarian)

REQ051. The librarian shall be able to view their read-only personal details, account status, membership role, and security status.

4.1.24. Log In (Student Assistant)

REQ052. The student assistant shall be able to log in using Microsoft O365 institutional credentials.

4.1.25. Dashboard (Student Assistant)

REQ053. The student assistant shall be able to view real-time statistics for daily visitors, total users, and active book loans.

REQ054. The student assistant shall be able to utilize quick shortcuts to the Circulation Desk, Attendance Logs, and Book Catalog modules.

REQ055. The student assistant shall be able to view a personal tracking section displaying their own current borrows, hold reservations, and recent attendance entries.

4.1.26. Catalog (Student Assistant)

REQ056. The student assistant shall be able to browse book cards displaying book titles, authors, cover images, and fractional availability indicators (e.g., available copies vs. total copies).

REQ057. The student assistant shall be able to search catalog books using a keyword bar (titles, authors, ISBNs) and refine results using category filters, availability statuses, and alphabetical sorting.

REQ058. The student assistant shall be able to click a "Reserve for Pickup" button on catalog items to place a hold.

4.1.27. Inventory (Student Assistant)

REQ059. The student assistant shall be able to view copy-level inventory items in a structured data table showing Title, Author, Category, ISBN, and generated book QR codes.

REQ060. The student assistant shall be able to filter inventory entries by category and keyword search.

4.1.28. Circulation Desk (Student Assistant)

REQ061. The student assistant shall be able to process front-desk borrowing and returning transactions using QR-code scanners

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4.1.29. My Borrowing (Student Assistant)

REQ062. The student assistant shall be able to view their personal borrow history table displaying resource metadata, borrow date, due date/return timeline, status tags, keyword search, and status filters (All, Active, Returned, Overdue).

4.1.30. Borrowing Logs (Student Assistant)

REQ063. The student assistant shall be able to access a view-only monitoring list of system-wide borrowing and returning transactions.

4.1.31. My Attendance (Student Assistant)

REQ064. The student assistant shall be able to view their personal attendance record displaying visit dates, check-in times, check-out times, session statuses, and an All-Time History view.

4.1.32. Attendance Logs (Student Assistant)

REQ065. The student assistant shall be able to record student check-ins and check-outs with timestamps by scanning library cards.

4.1.33. Notification (Student Assistant)

REQ066. The student assistant shall be able to receive alerts via header drop-down panels and filter/search updates on a dedicated notifications page.

4.1.34. My Profile (Student Assistant)

REQ067. The student assistant shall be able to view their read-only account profile and security details.

4.1.35. Log In and Profile Completion (Student)

REQ068. The student shall be able to log in using their Microsoft O365 institutional account.

REQ069. The student shall be required to complete a profile onboarding step (submitting residential address, contact numbers, and academic program) upon initial login.

REQ070. The system shall place new student accounts into a "Pending" status upon submission, restricting full feature access until administrator approval.

4.1.36. Dashboard (Student)

REQ071. The student shall be able to view a digital Library Card displaying their profile picture, name, program, academic year, scannable QR code, and Flip/Export card action buttons.

REQ072. The student shall be able to view personalized tracking sections for Current Borrows, Reservations (with cancel hold option), and Recent Attendance snapshots (check-in/out timestamps and visit durations).

REQ073. The student shall be able to access an interactive Support & FAQs panel to learn about library systems and policies.

4.1.37. Category (Student)

REQ074. The student shall be able to search and filter book collections, view fractional availability indicators, and place "Reserve for Pickup" holds.

4.1.38. My Borrow (Student)

REQ075. The student shall be able to view their transactional borrow timeline table showing resource details, borrow dates, due/return dates, keyword timeline search, and status filters (All, Active, Returned, Overdue).

4.1.39. My Attendance (Student)

REQ076. The student shall be able to view their personal check-in/check-out dates, entry/exit timestamps, session statuses, and complete chronological visit history.

4.1.40. Notification (Student)

REQ077. The student shall be able to receive alerts regarding library usage, event updates, and hold statuses via header bell notifications and dedicated alert pages.

4.1.41. My Profile (Student)

REQ078. The student shall be able to view their read-only account details including full name, system ID, official email, contact number, residential address, account status, membership role, and security status.

4.2. Non-Functional Requirements

4.2.1. Performance

REQ079. The system shall process and display real-time transaction updates (such as QR-based borrowing, returning, and entry/exit card scans) immediately upon scanning.

REQ080. The system shall maintain smooth response times for concurrent user activities (such as simultaneous catalog browsing, reservation holds, and administrative reporting) without performance degradation.

REQ081. The system shall load search query results, inventory filtering options, and analytics visualizations without noticeable interface lag.

4.2.2. Availability

REQ082. The system shall remain operationally accessible 24 hours a day, 7 days a week, except during planned platform maintenance windows coordinated with Microsoft O365 infrastructure.

4.2.3. Security

REQ083. The system must enforce user authentication exclusively through Microsoft O365 institutional accounts across all user roles (Super Admin, Librarian, Student Assistant, Student) to prevent unauthorized access.

REQ084. The system must enforce strict Role-Based Access Control (RBAC), ensuring that non-administrative accounts (Students) cannot view administrative tools, audit logs, or backend directories.

REQ085. The system must maintain an immutable security audit trail that records and protects all administrative activity and data modifications against unauthorized alteration.

4.2.4. Usability

REQ086. The system must feature an intuitive, visually clear, and consistent interface layout tailored specifically to each of the four distinct user roles.

REQ087. The system shall utilize consistent color-coded visual badges (e.g., green for Returned, blue/yellow for Active, red for Overdue) across all tables to optimize operational readability.

REQ088. The system shall provide quick-access keyword search bars and dropdown filters across all catalog, directory, log, and notification interfaces.

4.2.5. Maintainability

REQ089. The system architecture shall be modularly designed to allow updates to system rules, borrowing configurations, or academic program lists without disrupting core circulation services.

4.2.6. Portability and Deployment

REQ090. The system must function consistently across major modern web browsers, including Google Chrome, Mozilla Firefox, Microsoft Edge, and Apple Safari, without requiring special extensions.

REQ091. The system shall be fully accessible and responsive across multiple hardware devices and operating systems (Windows, macOS, Linux, iOS, Android).

REQ092. The system shall support customizable data export operations, allowing administrative catalog data and audit logs to be downloaded in Spreadsheet CSV, JSON, and ZIP formats.

TAB 3: Calendar of Activities

CALENDAR OF ACTIVITIES

MONTH

FEBRUARY

MARCH

APRIL

MAY

JUNE

JULY

AUGUST

SEPTEMBER

OCTOBER

NOVEMBER

DECEMBER

Activities

Requirements

Gathering

Title Proposal

Brainstorming

Researching

Documentation

Chapter 1:

Introduction

Chapter 2: Review

Related Literature

Chapter 3: Technical

Background

Chapter 4: Methodolog, Result and Discussion

Chapter 5: Conclusion and Recommendation

System Development

Transcript of

Interview

TAB 4: specific problem

Specific Problem

Super Admin

How to implement a secure login module using Microsoft O365 authentication for Super Administrator?

In this module, Super Administrator can access the system using their institutional accounts to verify their identity. This ensures that only authorized personnel can access sensitive library administrative tools and data through a standardized sign-in process.

How to develop an automated and detailed Inventory Module for library asset management?

In this module, super administrator and staff can efficiently manage the library's collection of books by utilizing an automated ISBN fetch feature to retrieve book information. It provides the necessary interface for staff to manually input specific organizational metrics—such as the Dewey Decimal number, book categories, number of copies, and exact shelf locations—ensuring precise tracking and optimized physical resource organization.

How to implement an automated Circulation Desk Module for efficient borrowing and returning processes?

In this module, the system utilizes QR-code scanning technologies to scan student IDs and book codes, automatically linking the borrower to the respective library item. This provides a streamlined workflow that drastically reduces manual data entry and minimizes transaction times for a more efficient counter service.

How to develop a comprehensive Borrowing Logs Module for tracking book transactions and statuses?

In this module, super administrator and staff can monitor a centralized, view-only list tracking when books are borrowed and returned. It displays all key transaction details including borrower identification, book data, checkout dates, and due timelines while utilizing a search bar and status filters (All, Active, Returned, Overdue) to allow for quick record retrieval and efficient oversight of overdue items.

How to design and implement an Analytics Module with visual data dashboards for tracking system trends and performance?

In this module, super administrator can filter historical operational data by specific timeframes such as 7 days, 30 days, or 1 year to monitor key metric summaries. By utilizing interactive line graphs for traffic and circulation trends, a donut chart for asset status breakdowns, and a leaderboard for high-demand books, the system provides a comprehensive visual tool for data-driven evaluation of library engagement and inventory performance.

How to establish a User Directory for user management and access control?

In this module, Super Administrator can manage the profiles and system access levels of all registered students. This ensures a secure database where library services are restricted to authorized individuals, allowing for efficient membership tracking and data protection.

How to implement an Attendance Logs Module for monitoring and tracking student traffic?

In this module, the super administrator can monitor daily library usage by scanning student library cards to track entry and exit events. The system displays a complete, real-time list of checked-in and checked-out individuals alongside exact timestamps, providing a streamlined process for recording and managing daily student traffic.

How to develop a comprehensive Setting and Policies Module for system rule configuration and operational maintenance?

In this module, super administrator can manage library rules and configurations through dedicated setup views to define circulation limits, program tracks, book categories, and announcements. It also integrates troubleshooting support logs alongside critical backend tools for batch account resets, automated email dispatches, and configuration testing, ensuring centralized control over system maintenance and policy alignment.

How to implement a secure Audit Logs Module for comprehensive activity tracking and system accountability?

In this module, a robust security tracking system records all system modifications and user activities in a detailed data table featuring timestamps, user identities, action types, and technical difference logs. Super administrators can efficiently navigate large datasets using a real-time search bar, specialized dropdown filters, and live data refreshes, while utilizing an export CSV feature to download historical logs for deeper security compliance and data analysis.

How to design and implement a dynamic Notification Module for real-time system alerts and updates?

In this module, super administrator are instantly alerted to critical system events and updates via a quick drop-down panel accessible from the system header. It provides a dedicated notifications page equipped with quick filter tabs and a targeted search bar, allowing staff to efficiently manage, find, and respond to specific alerts for enhanced situational awareness.

How to implement a view-only My Profile Module for secure account detail verification?

In this module, logged-in users can view a secure, centralized overview of their personal account details. It displays critical identity and contact information including the user's full name, system ID, official email, contact number, residential address, account status, membership role, and security status ensuring a straightforward way for individuals to verify their registered profile data without modifying system records.

Librarian

How to integrate a secure Login Module using Microsoft O365 authentication for Librarian?

In this module, Librarian can access the system using their institutional accounts to verify their identity. This ensures that only authorized personnel can access sensitive administrative tools through a standardized, secure sign-in process that eliminates manual credential management and enhances enterprise-level access control.

How to develop a Centralized Dashboard for library inventory monitoring?

In this module, Librarian can view a real-time overview of the library’s total book inventory and current stock status. This provides the necessary data for efficient resource monitoring and data-driven management,

enabling staff to make informed, immediate decisions regarding asset availability and collection scaling.

How to implement a Circulation Desk module utilizing QR-Code scanning?

In this module, Librarians can process book transactions by scanning QR codes for both student identification and book processing. This automates the link between the borrower and the item, significantly reducing manual data entry and transaction time to provide a faster, error-free checkout workflow.

How to develop an Attendance Module for monitoring student check-in and check-out?

In this module, Librarians can automate the tracking of library visitors using QR-code technology to log the specific times students check in and check out of the facility. This directly addresses the problem of manual logbooks, which are often inaccurate and time-consuming, by establishing an instant, reliable, and digital data-gathering process for daily facility usage.

How to develop a digital borrowed history module for administrative tracking?

In this module, Librarians can maintain and access a comprehensive digital record of all active, returned, and overdue books. This allows for the effective monitoring of borrowing trends and management of outstanding returns, replacing traditional tracking methods with a centralized system that improves transparency and accountability.

How to develop an Analytics Module for monitoring library usage and collection of books?

In this module, Librarians can view a consolidated summary of library activities, including daily visits, total collection count, and library user statistics. This directly addresses the problem of fragmented data by providing a centralized dashboard that visualizes book loan patterns and popular titles, enabling the admin to make informed, data-driven management decisions regarding inventory distribution and resource allocation.

How to establish a User Directory for student profile management?

In this module, Librarians can manage the profiles and system access levels of all registered students. This ensures a secure database where library services are restricted to authorized individuals, providing an organized verification interface that streamlines account oversight and protects sensitive student data.

How to develop a Settings and Policies module for system customization?

In this module, Librarians can configure borrowing limits, reservation rules, and academic categories. This allows the system to be tailored to specific institutional policies and operational requirements, ensuring that automated platform constraints perfectly align with the changing logistical needs of the facility.

How to develop a Notification Module for library Librarians?

In this module, Librarians will check for unread alerts and read specific details of system activities. This directly addresses the problem of missed tasks by allowing the admin to navigate directly to relevant modules if an action is required or dismiss notifications after review, ensuring timely administrative responses and a highly responsive workflow.

How to develop a Profile Module for Librarians to manage their identity?

In this module, Librarians can access their account details to ensure their records are correct. This addresses the need for profile personalization by allowing users to update their profile photo, providing a reliable way to maintain an up-to-date visual identification within the system while keeping other primary personal data secure and view-only.

Student Assistant

How to integrate a secure login module using Microsoft O365 authentication for student assistants?

In this module, Student Assistants use their institutional accounts to sign in and verify their identity. This ensures that only verified students gain access to the library's digital services, utilizing a standardized institutional single sign-on process to maintain platform security while providing seamless access for student workers.

How to develop a Centralized Dashboard for library inventory monitoring?

In this module, Student Assistants can view a real-time overview of the library’s total book inventory and current stock status. This provides the necessary data for efficient resource monitoring and data-driven management, enabling student workers to quickly check asset availability and support the library's daily operational tasks.

How to provide a personalized Student Assistants Dashboard for account management?

In this module, Student Assistants can view their digital library cards and track their personal borrowing status. This portal provides a convenient way for student workers to manage their own accounts and stay updated on their current transactions, ensuring easy access to identification and real-time visibility over their personal lending history.

How to develop a searchable Catalog module with real-time availability?

In this module, Student Assistants can search for books and see their real-time availability status online. This improves accessibility by allowing students to browse and check the current status of the collection without physically visiting the library premises, streamlining the research process and reducing unnecessary trips to the physical book stacks.

How to develop an Inventory Module for student assistants to manage library collections?

In this module, student assistants can access and manage the digital record of library books. This directly addresses the problem of manual data entry and inefficient inventory tracking by providing tools to add new resources through QR-code scanning or ISBN fetching, ensuring that the library’s collection records are updated accurately, rapidly, and efficiently.

How to develop an Attendance Module for monitoring student check-in and check-out?

In this module, administrators can automate the tracking of library visitors using QR-code technology to log the specific times students check in and check out of the facility. This directly addresses the problem of manual logbooks, which are often inaccurate and time-consuming, by establishing an automated, reliable digital log that minimizes human error and streamlines daily visitor tracking.

How to maintain a Student Borrowing History module for personal tracking?

In this module, administrators can maintain and access a comprehensive digital record of all active, returned, and overdue books. This allows for the effective monitoring of borrowing trends and the streamlined management of outstanding returns, replacing decentralized records with an organized historical database that ensures operational accountability and precise tracking.

How to develop a Notification Module for student assistants?

In this module, student assistants will be able to view unread alerts regarding their library transactions. This directly addresses the need for clear communication by allowing student assistants to act promptly on reminders, such as overdue books, or simply mark informative alerts as read before returning to their dashboard, ensuring they remain aware of their account status and deadlines.

How to develop a Profile Module for student assistants to manage their identity?

In this module, student assistants can access their account details to ensure their records are correct. This addresses the need for profile personalization by allowing users to update their profile photo, providing a

secure way to maintain an up-to-date visual identification within the system while keeping other primary personal data secure and view-only to maintain system data integrity.

Student

How to integrate a secure login module using Microsoft O365 authentication for students?

In this module, students use their institutional accounts to sign in and verify their identity. This ensures that only verified students gain access to the library's digital services, utilizing a standardized institutional single sign-on process to protect user privacy while providing seamless, authenticated access to the library platform.

How to implement a standardized user profile completion and administrative verification process?

In this module, new users must provide essential information, including their address, contact number, and academic program. This ensures that the system maintains accurate records for every student and provides a manual verification layer where administrators must approve accounts before full system access is granted, preventing unauthorized or incomplete profiles from accessing restricted library services.

How to provide a personalized Student Dashboard for account management?

In this module, students can view their digital library cards and track their personal borrowing status. This portal provides a convenient way for students to independently manage their accounts and stay updated on their current transactions, ensuring easy access to digital identification and real-time visibility over their active book loans and return deadlines.

How to develop a searchable Catalog module with real-time availability?

In this module, students can search for books and see their real-time availability status online. This improves accessibility by allowing students to browse and check the current status of the collection without physically visiting the library premises, streamlining their research process and allowing them to plan their book loans efficiently.

How to develop a module for students to monitor their personal library attendance records?

In this module, students can access a digital log of their entry and exit history. This directly addresses the problem of students lacking visibility into their own library usage, providing them with a transparent record of all recorded "time-in" and "time-out" sessions to help them track their study hours and facility visits while ensuring accountability regarding their library attendance data.

How to maintain a Student Borrowing History module for personal tracking?

In this module, students can view their history of active, returned, and overdue books. This provides clear visibility of personal transaction history and return deadlines, directly helping students stay organized, manage their active loans responsibly, and avoid penalties by tracking outstanding items independently.

How to develop a Notification Module for students?

In this module, students will be able to view unread alerts regarding their library transactions. This directly addresses the need for clear communication by allowing students to act promptly on critical reminders, such as overdue books, or simply mark informative alerts as read before returning to their dashboard, keeping them proactively informed about their account standing.

How to develop a Profile Module for students to manage their identity?

In this module, students can access their account details to ensure their records are correct. This addresses the need for profile personalization by allowing users to update their profile photo, providing a secure way to maintain an up-to-date visual identification within the system while keeping other primary personal data

secure and view-only to prevent unauthorized alterations to institutional student records.

TAB 5: Design of Software

Design of Software, System, Product, and/or Processes

In designing our software, system, and process, we utilized structured architectural and design approaches to create a secure, efficient, and user-centered system. The system follows a layered architecture consisting of the Presentation, Process/Logic, Database Client, and Database Core layers, ensuring clear separation of responsibilities and maintainability. Database design applies Third Normal Form (3NF) to minimize redundancy and maintain data integrity, while ACID transactions, row-level locking, advisory locks, and database triggers support reliable and concurrent library operations. Security was incorporated through authentication, role-based access control (RBAC), input validation using Zod, Row Level Security (RLS), restricted database procedures, and comprehensive audit logging. The system also applies ISO/IEC 25010 quality attributes, focusing on functional suitability, performance efficiency, compatibility, usability, reliability, security, maintainability, and portability. Integrating these architectural, database, security, and quality design practices resulted in a robust, scalable, and user-friendly library management product.

Development and Testing process

The researchers developed the capstone project by using Visual Studio Code (VS Code) as it emerged as the preferred Integrated Development Environment (IDE) due to its features and flexibility; its customization options and integrated tools make it an ideal choice for developing system.

Web Development with VS Code

VS Code improved the development workflow by making coding and debugging easy. The system was built around the integrated capabilities of the platform and its large extension ecosystem.

Capstone Adviser’s Guidance

Development and testing was helped considerably by the researcher’s capstone adviser approach involvement. The researchers held regular meetings and consultations, gathering clear feedback to pinpoint the necessary improvements and modifications to ensure the project met the highest standards and stayed on track with its core objectives.

Preliminary Design

For web development Visual Studio Code is used in the initial design.While the backend uses TypeScript alongside SQL (PostgreSQL) for database management and application logic, the frontend combines HTML, CSS, and TSX for user interface layout and styling.

LOGIN

[image omitted][image omitted]

SUPERADMIN SIDE

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STUDENT SIDE

[image omitted][image omitted][image omitted][image omitted][image omitted][image omitted][image omitted]

LIBRARIAN SIDE

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STUDENT ASSISTANT SIDE

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MOBILE VIEW

[image omitted][image omitted][image omitted]

[image omitted][image omitted][image omitted]

TAB 6: Result and Discussion

Testing

In this part, the proponents present the testing activities conducted for Lumina: A Web-Based Library Book Reservation, Borrowing, and Inventory Management System with QR-Based Scanning for STI College Alabang. Testing focuses on verifying whether the developed modules and functions operate according to their intended requirements. The system follows the Iterative Development method, where each feature is tested after implementation to identify errors and necessary improvements. The testing includes Microsoft O365 authentication, book searching and reservation, QR-based scanning, borrowing and returning, attendance monitoring, inventory management, notifications, and administrative functions, with the results used to improve the system in the next development cycle.

Description of Prototype

The prototype is a web-based library management system designed to provide students and library personnel with a centralized platform for accessing and managing library services. It includes features such as book searching and reservation, borrowing and returning, inventory management, attendance monitoring, notifications, and user profile management. The system uses Next.js, TypeScript, Supabase, Tailwind CSS, and HTML5-QRCode to support its web interface, database, and QR-based scanning functions. Overall, the prototype is designed to improve the efficiency of library transactions, organize book records, and provide convenient access to library services for students and library personnel.

Implementation Plan

The implementation of A Web-Based Library Book Reservation, Borrowing, and Inventory Management System with QR-Based Scanning for STI College Alabang resulted in a centralized platform that supports book searching, reservations, borrowing and returning, inventory management, and attendance monitoring. Students can search and reserve books, view borrowing history and due dates, check attendance records, receive notifications, and access their profiles. Library personnel can manage book records and transactions using QR-based scanning, while administrators can manage inventory, borrowing logs, analytics, users, settings, and audit logs. Overall, the implemented features provide an organized digital platform that supports library operations, improves access to information, and reduces reliance on manual recording.

Implementation Results

The implementation of A Web-Based Library Book Reservation, Borrowing, and Inventory Management System with QR-Based Scanning for STI College Alabang resulted in a functional system that integrates the major library processes into one centralized web-based platform. The implemented modules allow students to browse available books, reserve and borrow books, view borrowing history, and manage their profiles, while administrators can manage book records, monitor inventory, process borrowing and returns, and manage user transactions. The QR-based scanning feature was also implemented to support faster and more accurate identification of books and borrowing transactions. Overall, the implementation results demonstrate that the developed system successfully addresses the identified limitations of the existing manual processes by providing a more organized, accessible, and efficient approach to library management.

TAB 7: Abstract

ABSTRACT

Title of research:

Researchers: Dadea, Jerico B.

Floriano, Kayle Andrew P.

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Resurreccion, Rhed R.

Tolentino, Keneth James B.

Degree: Bachelor of Science in Information Technology

Date of Completion: May 2026

Keywords: Library Book Reservation, Borrowing, Inventory Management, QR-Based Scanning

Lumina, a library book reservation, borrowing, and inventory management system are crucial aspects of academic administration in STI College Alabang, ensuring that educational institutions operate efficiently while meeting the needs of the faculty and students. This capstone project aims to explore the challenges and solutions in creating optimized tracking that balance book availability, borrowing queues, inventory counts, and conflict resolutions.

The study also evaluates QR-based scanning integration to prevent manual data entry errors while ensuring a balanced distribution of library resources.

By leveraging web hosting and securing a dedicated domain, the project aims to establish a widely and secure platform for day-to-day operations. The adoption of a web base application framework aligns with institution standards, providing a polished and professional interface for all users.

The primary objective is to offer a system that supports strategic tasks, improves utilization, enhances student learning experience, and addresses the academic needs of students.

TAB 8: Conclusion and Recommendation

CONCLUSION

The development and implementation of Lumina directly addresses the longstanding inefficiencies associated with manual library operations at STI College Alabang. By replacing time-consuming paper logbooks and manual shelf checks with a centralized, web-based platform, the project unifies book reservation, borrowing, circulation, and inventory auditing into a cohesive digital ecosystem.

Through the integration of QR-based scanning technologies, Microsoft O365 enterprise authentication, and a flexible iterative development methodology, Lumina ensures real-time data accuracy, minimizes administrative wait times, and safeguards institutional records. The modular system design fosters an intuitive, user-centric environment that simplifies daily tasks for administrators, librarians, student assistants, and students alike. Ultimately, transitioning from traditional record-keeping to a modern, cloud-based digital framework resolves immediate operational bottlenecks while laying a scalable, future-ready foundation. This transformation directly supports STI College Alabang's broader commitment to academic excellence, data integrity, and digital innovation in library management.

Recommendation The researchers conducted a complete evaluation of the system and identified advanced areas for system expansion, specifically AI-Driven Predictive Shelf-Life and Condition Forecasting, Automated Data-Anonymized CSV/JSON Audit Exporters for Accreditation, and Offline Queueing for Mobile QR Scanning .

The proponents propose the following recommendation: 1.AI-Driven Predictive Shelf-Life and Condition Forecasting The integration of a machine learning regression model within the management information system (MIS) will optimize physical asset preservation. By tracking how often a specific physical book is circulated, its age, and historical damage logs, the system can flag textbooks that are at high risk of structural wear, allowing librarians to proactively plan for repairs, rebinding, or replacements before the material becomes unusable.

2.Automated Data-Anonymized CSV/JSON Audit Exporters for Accreditation

Implementing an automated compliance module within the super-admin panel will greatly simplify institutional evaluation processes. The system can generate encrypted, data-anonymized reports in both CSV and JSON formats that demonstrate precise library utilization metrics, book-to-student ratios, and inventory turn rates over specific academic years without compromising student privacy or data integrity controls.

3. Queueing for Mobile QR Scanning.

Give librarians and student assistants who perform inventories in remote areas of the library or have poor internet connectivity offline access so they can continue scanning book QR codes; once a stable internet link is established, the scanned data and transaction logs will be automatically synchronized with the Supabase database.

TAB 9: Input and Output of the System

INPUT/OUTPUT/REPORTS

Input Table

The inputs refer to the data and information provided by the students and library personnel to the Lumina system. These inputs are used by the system to process library transactions, manage book records, and maintain user information.

1. Student Information

Field

Description

Example

Student Name

Name of the student using the system

Juan Dela Cruz

Student ID

Unique identification number of the student

02000123456

STI Email

Official email account used for authentication

delacruz.123456@alabang.sti.edu.ph

Program

Student's academic program

BS Information Technology

Year Level

Current year level of the student

3rd Year

2. Book Information

Field

Description

Example

Book Title

Title of the library book

Introduction to Programming

Author

Author of the book

John Smith

ISBN

International Standard Book Number

978-1234567890

Category

Classification of the book

Computer Science

Quantity

Number of available copies

5

Book Status

Current availability of the book

Available

3. Reservation Information

Field

Description

Example

Student ID

Student making the reservation

2026-12345

Book ID

Unique identifier of the selected book

BK-001

Reservation Date

Date when the reservation was made

September 29, 2026

Reservation Status

Current status of the reservation

Pending

4. Borrowing and Return Information

Field

Description

Example

Student ID

Student borrowing or returning the book

2026-12345

Book ID

Identifier of the borrowed book

BK-001

QR Code

QR code assigned to the book or library transaction

QR-BK001

Borrow Date

Date when the book was borrowed

September 29, 2026

Due Date

Date when the book should be returned

October 6, 2026

Return Date

Date when the book was returned

October 5, 2026

5. Inventory Information

Field

Description

Example

Book ID

Unique identifier of the book

BK-001

Book Title

Title of the book

Introduction to Programming

Category

Category of the book

Computer Science

Quantity

Number of copies recorded in the inventory

5

Availability

Current availability of the book

Available

Output Table

The outputs are the results generated by the Lumina system after processing the inputs. These outputs provide students and library personnel with information regarding books, reservations, borrowing transactions, and inventory.

Output Field

Description

Output Type

Book Search Results

List of books matching the student's search criteria

System Display

Book Details

Information about a selected book, including title, author, category, and availability

System Display

Reservation Confirmation

Confirmation and status of a student's book reservation

System Display / Notification

Borrowing Record

Record containing the student, book, borrow date, and due date

Table View

Return Record

Record of returned books and their return date

Table View

Borrow History

List of previous and current borrowing transactions of a student

Table View

Updated Book Status

Current availability status of a book after reservation, borrowing, or return

System Display

Updated Inventory

Updated records of books and their available quantities

Table View

QR Scan Result

Identified student or book information obtained from the scanned QR code

System Display

Notifications

System messages regarding reservations, borrowing, returns, or other library activities

Notification

Student Library Card

Digital library card containing the student's library information

Digital Card

Report Specifications

The reports generated by the Lumina system are intended to support library personnel in monitoring library transactions, managing inventory, and maintaining accurate records.

Report Name

Description

Report Type

Audience

Borrowing History Report

Contains records of books borrowed by students, including borrowing and due dates

Table / Printable

Library Personnel

Return Records Report

Contains information about returned books and their corresponding return dates

Table / Printable

Library Personnel

Inventory Report

Provides a summary of library books, quantities, and availability status

Table / Printable

Library Personnel

Reservation Report

Contains records of book reservations and their current status

Table / Printable

Library Personnel

User Directory Report

Contains registered student information and account details

Table View

Library Personnel

Book Records Report

Provides information about books registered in the library system

Table / Printable

Library Personnel

Transaction Summary Report

Provides a summary of borrowing, returning, and reservation activities

Table / Printable

Library Personnel

The Input/Output section demonstrates how information is entered into the Lumina system and how the system processes and presents the resulting information. The inputs provided by students and library personnel allow Lumina to perform its library management functions, while the outputs provide updated records, confirmations, notifications, and reports for the appropriate users.

TAB 10: Flowcharts


[image omitted]

Figure 3 Entry Authentication flowchart


[image omitted]

Figure 4 Super-Admin Dashboard Flowchart


[image omitted]

Figure 4.1 Super-Admin Inventory Flowchart


[image omitted]

Figure 4.2 Super-Admin Circulation Desk Flowchart


[image omitted]

Figure 4.4 Super-Admin Borrowing Logs Flowchart

[image omitted]

Figure 4.5 Super-Admin Analytics Flowchart

[image omitted]

Figure 4.6 Super-Admin User Directory Flowchart

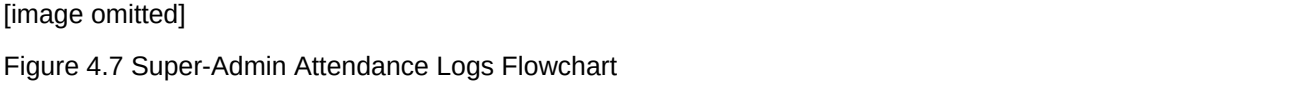

[image omitted]

Figure 4.7 Super-Admin Attendance Logs Flowchart

[image omitted]

Figure 4.8 Super-Admin Setting & Policies Flowchart

[image omitted]

Figure 4.9 Super-Admin Audit Logs Flowchart

[image omitted]

Figure 4.10 Super-Admin Profile Flowchart

[image omitted]

Figure 4.11 Super-Admin Notification Flowchart

[image omitted]

Figure 5 Librarian Dashboard Flowchart

[image omitted]

Figure 5.1 Librarian Inventory Flowchart

[image omitted]

Figure 5.2 Librarian Circulation Desk Flowchart

[image omitted]

Figure 5.3 Librarian Borrowing Logs Flowchart

[image omitted]

Figure 5.4 Librarian Analytics Flowchart

[image omitted]

Figure 5.5 LIbrarian User Directory Flowchart

[image omitted]

Figure 5.6 Librarian Attendance Logs Flowchart

[image omitted]

Figure 5.7 Librarian Settings & Policies Flowchart

[image omitted]

Figure 5.8 Librarian Profile Flowchart

[image omitted]

Figure 5.9 Librarian Notification Flowchart

[image omitted]

Figure 6 Student-Assistant Dashboard Flowchart

[image omitted]

Figure 6.1 Student-Assistant Inventory Flowchart

[image omitted]

Figure 6.2 Student-Assistant Catalog Flowchart

[image omitted]

Figure 6.3 Student-Assistant Circulation Desk Flowchart

[image omitted]

Figure 6.4 Student-Assistant My Attendance Flowchart

[image omitted]

Figure 6.5 Student-Assistant Attendance Logs Flowchart

[image omitted]

Figure 6.6 Student-Assistant My Borrow Flowchart

[image omitted]

Figure 6.7 Student-Assistant Borrowing Logs

[image omitted]

Figure 6.8 Student-Assistant Profile Flowchart

[image omitted]

Figure 6.9 Student-Assistant Notification Flowchart

[image omitted]

Figure 7 Student Dashboard Flowchart

‘;

[image omitted]

Figure 7.1 Student Catalog Flowchart

[image omitted]

Figure 7.2 Student My Borrow Flowchart

[image omitted]

Figure 7.3 Student My Attendance Flowchart

[image omitted]

Figure 7.4 Student Profile Flowchart

[image omitted]

Figure 7.5 Student Notification Flowchart

TAB 11: Tables

Resources

Specification:

Hardware Requirements

Processor

Intel Core i5-12400

Graphics Card

AMD Radeon RX 580

Memory

32 GB

Disk Storage

500 GB

Table 1 Hardware Requirements

Software Requirements

Operating System

Windows 11

Integrated Development Environment (IDE) and/or Code Editors

Visual Studio Code

Database

Supabase

Programming Languages and other Technologies

Next.js, TypeScript,

Documentation

Microsoft 0365 Word

Table 2 Software Requirements

Requirements Analysis

Minimum Hardware Requirements (For Development):

MINIMUM

RECOMMENDED

Processor

Intel Core i3-10100 @ 3.6- GHz (4 cores, 8 threads)/ AMD Ryzen 3 3200G @ 3.6GHz (4 cores, threads)

Intel Core i5-11400 @ 2.60GHz (6 cores, 12 threads) / AMD Ryzen 5 5600X

Memory

8GB DDR4

3.7GHz (6 cores, 12 threads) 16GB DDR4

Graphics Card

Integrated Graphics (e.g., Intel UHD Graphics 630 / AMD Radeon Vega 8)

Dedicated Graphics (e.g., NVIDIA GeForce GTX 1660 6GB / AMD Radeon RX 5600 XT 6GB)

Disk Storage

128GB

256GB or more

Table 3 Minimum Hardware Requirements (For Development)

Minimum Hardware Requirements (For Users):

Processor

Intel Core 2 Duo E8400 @ 3.0GHz / AMD Athlon II X2 250 @3.0GHz.

Memory

4GB RAM

Graphics

Integrated sound card (100% DirectX 10 compatible)

Table 4 Minimum Hardware Requirements (For Users)

TAB 12: User Guide

USERS GUIDE

This guide provides instructions for students and library personnel on how to navigate and efficiently use the Lumina system. The system is designed to simplify library book reservation, borrowing, returning, inventory management, and attendance monitoring through a centralized web-based platform.

Lumina uses Microsoft 365 authentication for secure user login and provides role-based access to features intended for students and authorized library personnel.

System Requirements

A device with internet access such as a PC, laptop, or smartphone.

A valid STI Microsoft 365 account.

A compatible web browser such as Microsoft Edge, Google Chrome, or Mozilla Firefox.

An active internet connection.

1. Logging Into the System

1.1 Launch the System

Open a compatible web browser and navigate to the Lumina system website.

1.2 Microsoft 365 Login

On the login page, select Sign in with Microsoft 365.

1.3 Enter Credentials

Enter the user's STI Microsoft 365 account credentials and complete the required authentication process.

1.4 Role-Based Access

After successful login, the system displays the dashboard and features according to the user's assigned role.

2. User Roles and Features

A. Students

Students use Lumina to search for books, make reservations, monitor borrowed books, and view their library information.

Key Features

View the Student Dashboard.

Search and browse available books.

View book details.

Reserve available books.

Report missing books.

View active borrowed books.

View reservations.

View borrowing history.

Access the digital library card.

Export the library card.

View frequently asked questions.

Manage profile information.

How To:

Search for a Book:

Navigate to the Catalog.

Enter the book title, author, or other available search information.

Select the desired book.

Review the book details and availability.

Reserve a Book:

Open the Catalog.

Select an available book.

Click Reserve.

Confirm the reservation.

View Active Borrowed Books:

Open the Student Dashboard.

Locate the Active Borrows section.

Review the currently borrowed books and their corresponding information.

View Borrowing History:

Navigate to Borrow History.

Review previously borrowed and returned books.

B. Library Personnel / Administrator

Library personnel are responsible for managing library resources, book circulation, users, inventory, and system settings.

Key Features

Manage library books and assets.

Add and edit book records.

Monitor book statuses.

Generate and print QR codes.

Process book checkouts.

Process book returns.

Scan QR codes.

Manage reservations.

Monitor borrowing records.

Manage users.

Configure library policies.

Manage book categories and programs.

Manage announcements.

How To:

Add a Book:

Navigate to the Admin Dashboard.

Select Add Item.

Enter the required book information.

Save the book record.

Edit a Book:

Open the Admin Dashboard.

Select the desired book or asset.

Click Edit Asset.

Update the required information.

Save the changes.

Generate or Print a QR Code:

Open the book or asset record.

Select Print QR.

Review the generated QR code.

Print the QR code and attach it to the corresponding book.

3. Circulation Desk Functions

The Circulation Desk is used by authorized library personnel to process book borrowing and returning transactions.

A. Checking Out a Book

Navigate to the Circulation Desk.

Select Checkout.

Launch the QR scanner.

Scan the student's library card or required identification.

Scan the QR code of the book.

Review the transaction details.

Select Process to complete the checkout.

B. Returning a Book

Navigate to the Circulation Desk.

Select Return.

Launch the QR scanner.

Scan the book's QR code.

Review the borrowing information.

Select Process to complete the return.

C. Resetting the Scanner

If a scanning transaction needs to be cleared, select Reset before starting a new transaction.

4. Managing Inventory

The inventory management functions allow authorized library personnel to monitor and maintain library book records.

Key Functions

Add new library assets.

Edit existing asset information.

Monitor book status.

Assign book categories.

Generate QR codes.

Update inventory records.

How To:

Navigate to the Admin Dashboard.

Select the appropriate inventory function.

Search for the required book or asset.

Add or update the necessary information.

Save the changes.

5. Managing Reservations

Library personnel can monitor and manage student book reservations.

How To:

Access the appropriate reservation or circulation section.

Review the list of active reservations.

Check the student's reservation information.

Verify the corresponding book.

Process the reservation according to the library policies.

6. Borrow History

The Borrow History section allows users to review borrowing transactions.

Students

Students can view their previous borrowing records, including books that have already been returned.

Library Personnel

Authorized personnel can review borrowing transactions to monitor library circulation and maintain accurate records.

7. User Directory

The User Directory allows authorized library personnel to manage registered users.

Key Functions

View registered users.

Search for users.

Review user information.

Manage user records according to their authorized access.

8. System Settings and Policies

The Settings section allows authorized library personnel to configure library policies and system information.

Available Settings

Borrowing window.

Maximum books allowed per student.

Renewal settings.

Reservation shelf life.

Queue slot limit.

Programs.

Book categories.

Library announcements.

How To:

Navigate to Settings.

Select the policy or configuration that needs to be updated.

Enter the required information.

Review the changes.

Save the updated settings.

9. Common Functions

A. Reporting a Missing Book

Students can report a book that cannot be located in the library.

Navigate to the Catalog.

Search for the missing book.

Open the book details.

Select Report Missing Book.

Submit the report.

B. Viewing the Library Card

Open the Student Dashboard.

Locate the Library Card section.

View the digital library card.

Select Export if a copy is needed.

C. Viewing FAQs

Open the Student Dashboard.

Navigate to FAQs.

Select the question related to the user's concern.

Review the provided information.

10. Updating Personal Information

Users can access their profile to review or update available personal information.

Navigate to Profile.

Review the displayed information.

Update the permitted information.

Save the changes.

11. Logging Out

To securely exit Lumina:

Click the user's profile icon.

Select Logout.

The system will end the current session.

Users should always log out after using Lumina on shared or public computers.

12. Troubleshooting and Support

Issue

Suggested Solution

Unable to log in

Check the Microsoft 365 account and internet connection.

Cannot access a feature

Check whether the feature is available for the user's assigned role.

Book cannot be reserved

Check the book's current availability and reservation status.

QR code cannot be scanned

Ensure the QR code is visible and properly positioned in front of the scanner.

Book information is incorrect

Report the issue to authorized library personnel.

Borrowing or returning transaction cannot be completed

Review the transaction details and try the process again.

System is not loading

Check the internet connection and refresh the browser.

For unresolved issues, users should contact the authorized library personnel or system administrator.

13. Quick Reference Table for Roles and Access

Role

Key Features

Student

Search books, view book details, reserve books, view active borrows, view borrowing history, view library card, report missing books, access FAQs, manage profile

Library Personnel / Administrator

Manage books, inventory, QR codes, reservations, borrowing and returning transactions, users, and system settings

14. Important Reminders

Users must log in using their authorized Microsoft 365 account.

Students should verify book information before making a reservation.

Library personnel should verify transactions before processing checkouts and returns.

QR codes should correspond to the correct library assets.

Users should log out after completing their session.

Library policies configured in Lumina should be followed when borrowing, returning, and reserving books.